Markov Decision Processes and Reinforcement Learning

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November 4, 2020

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1. Markov Decision Processes
2. Epistemic Uncertainty in MDPs
3. Reinforcement Learning
4. My Contribution

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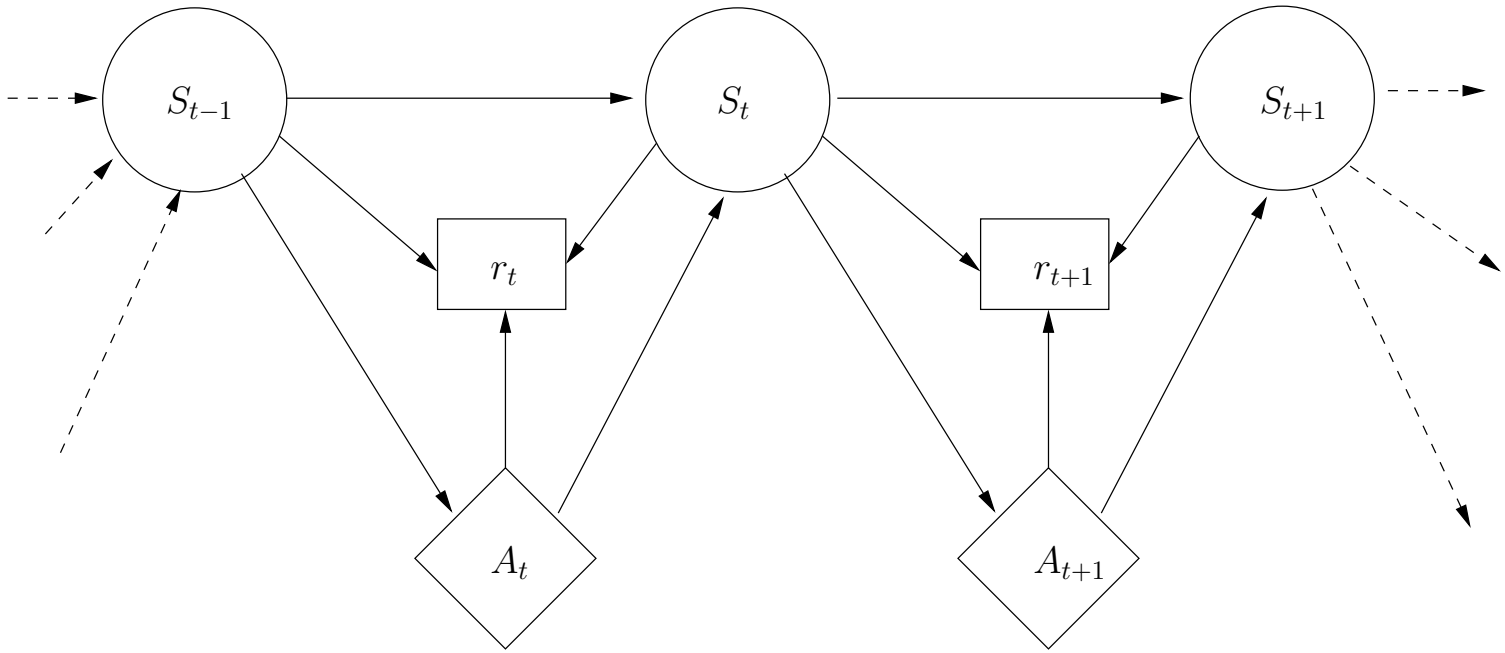
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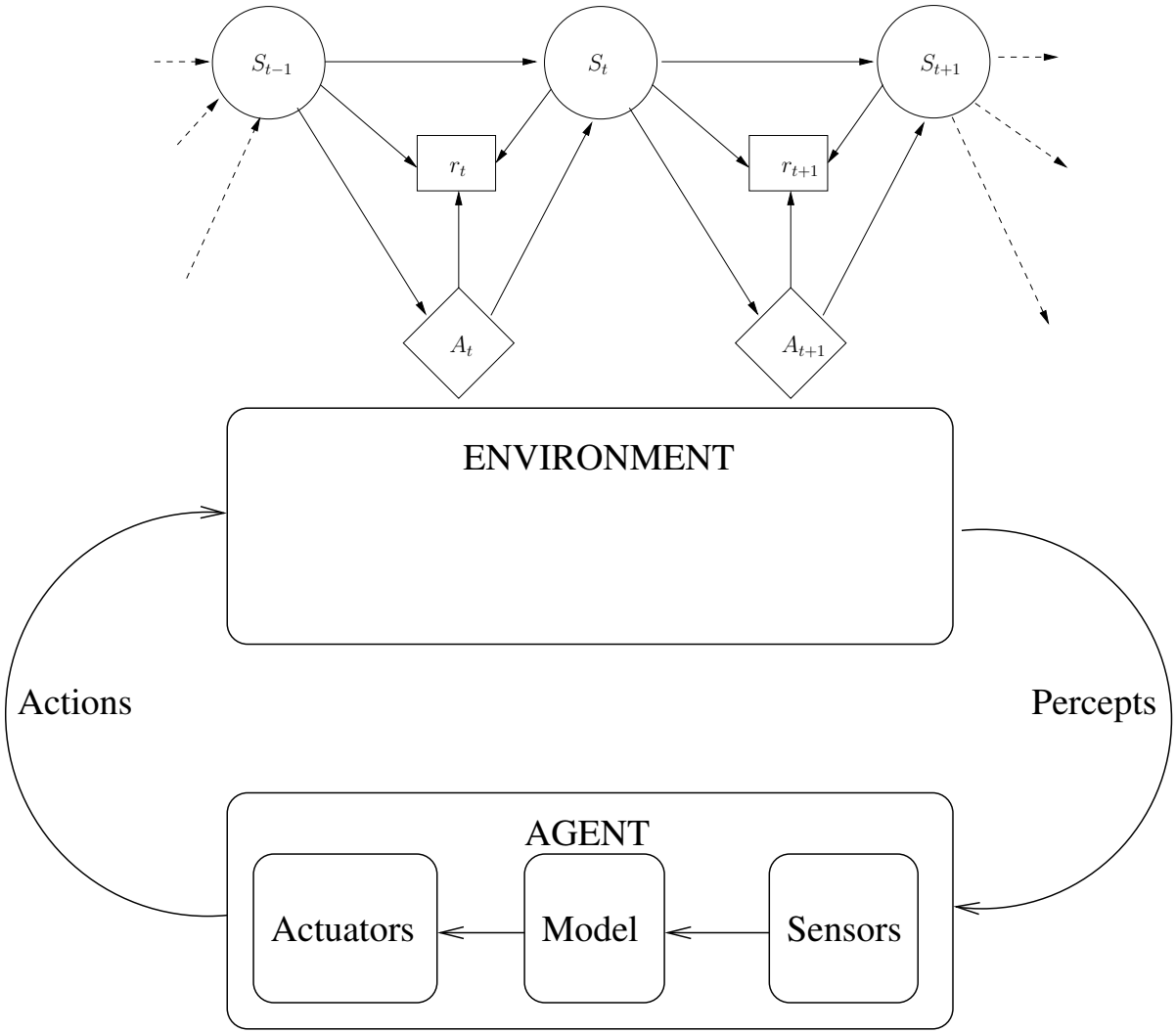
MDP Model



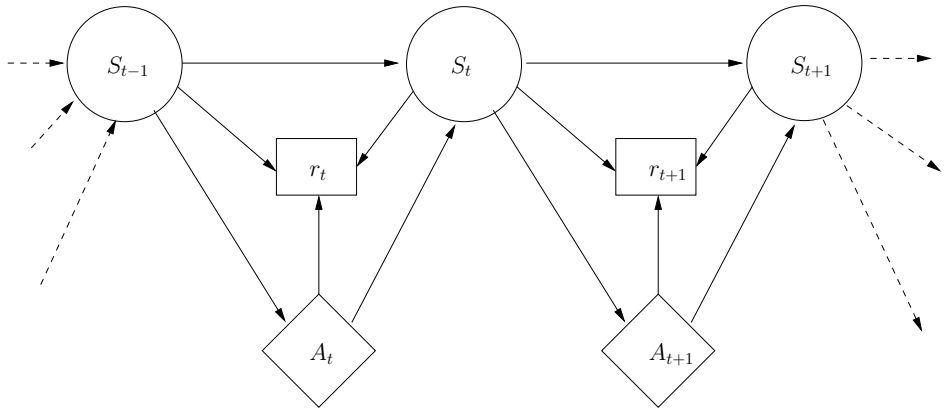
$$\langle \mathcal{S}, \mathcal{A}, T, r, s_0 \rangle$$

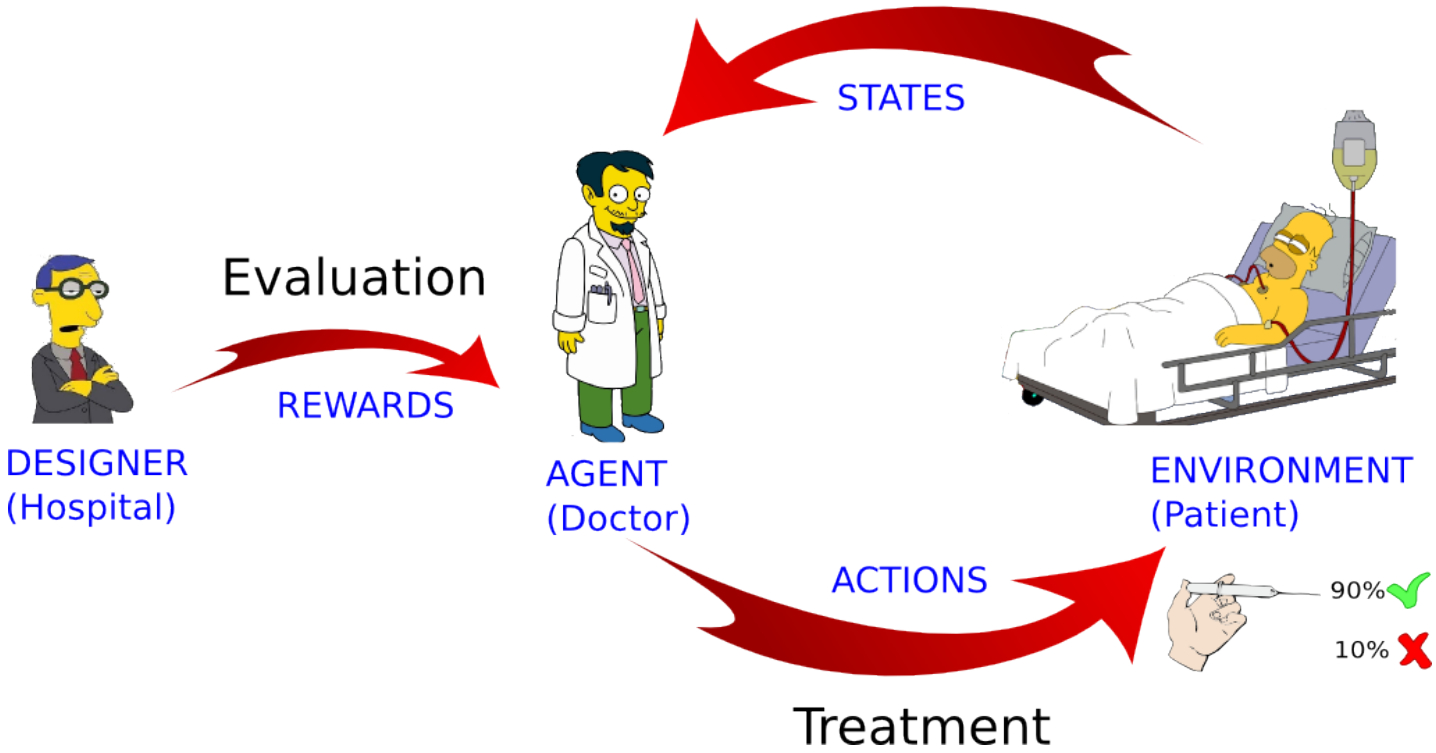
- $\mathcal{S}$: state space
- $\mathcal{A}$: action space
- $T(s, a, s')$: stochastic transition function ($Pr(s'|s, a)$)
- $r(s, a, s')$: reward function (scalar function)
- s_0: initial state

MDP Model

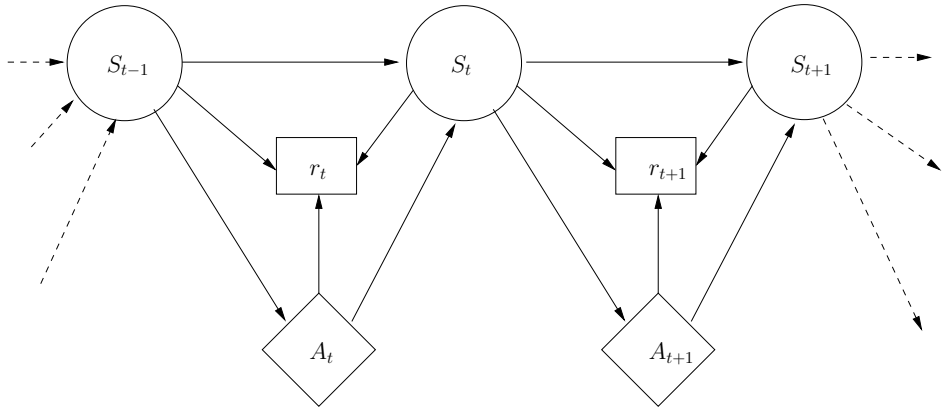

S_{t-1}
S_t
S_{t+1}
r_t
r_{t+1}
A_t
A_{t+1}
ENVIRONMENT
Actions
Percepts
AGENT
Actuators
Model
Sensors

MDP Model


S_{t-1}
S_t
S_{t+1}
r_t
r_{t+1}
A_t
A_{t+1}

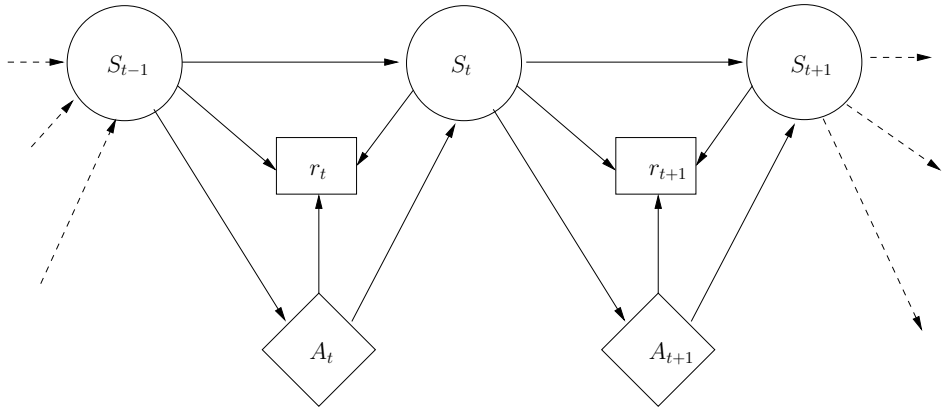

Disease
STATES
Evaluation
REWARDS
DESIGNER
(Hospital)
AGENT
(Doctor)
ENVIRONMENT
(Patient)
ACTIONS
90%
10%
Treatment

Policies



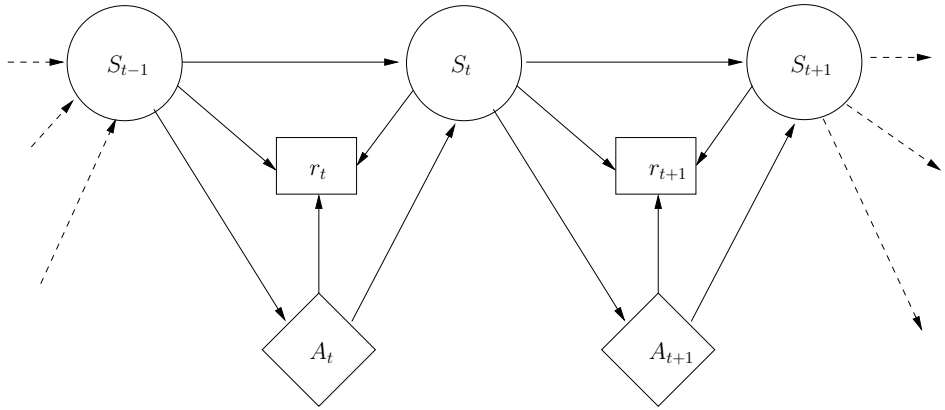
- How to plan?

Policies



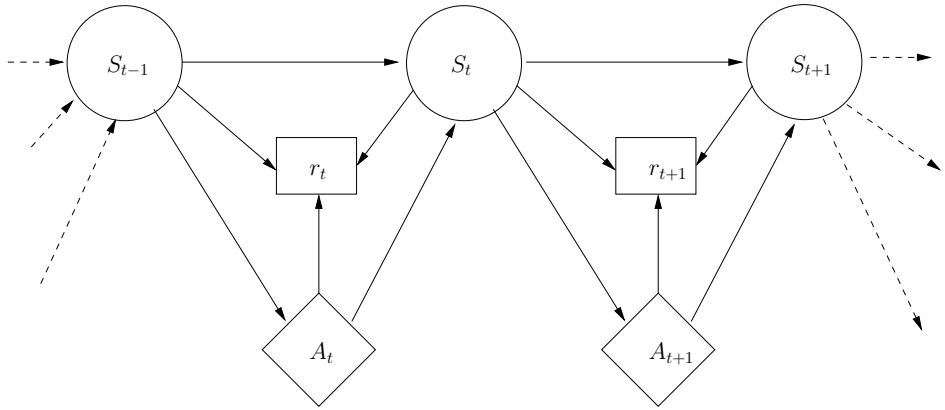
- How to plan? A sequence of actions is useless

Policies



- How to plan? A sequence of actions is useless
- Conditional policies, $\pi : \mathcal{S} \mapsto \mathcal{A}$

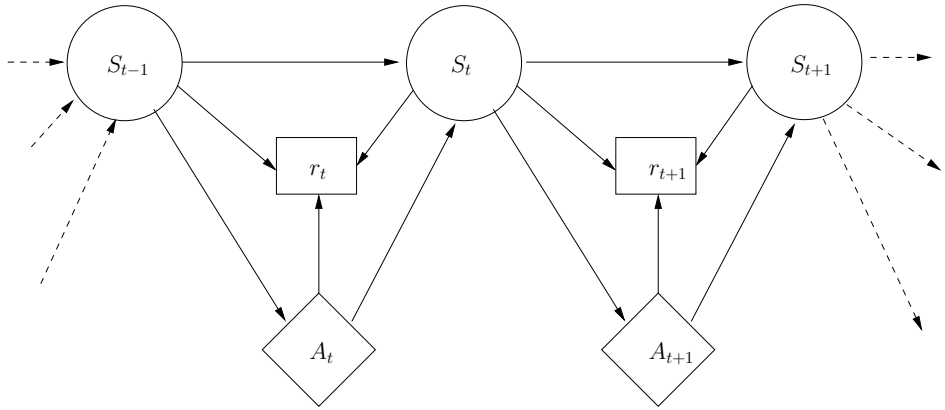
Policies



- How to plan? A sequence of actions is useless
- Conditional policies, $\pi : \mathcal{S} \mapsto \mathcal{A}$
- Policy evaluation:

$$V_H^\pi = \mathbb{E}\left[\sum_{t=0}^{H} r(S_t, \pi_t(S_t), S_{t+1}) \middle| \pi, s_0\right]$$

Policies



- How to plan? A sequence of actions is useless
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- Policy evaluation:

$$V_H^\pi = \mathbb{E}\left[\sum_{t=0}^{H} r(S_t, \pi_t(S_t), S_{t+1}) \middle| \pi, s_0\right]$$

- Value Function:

$$V_t^\pi(s) = \sum_{s' \in \mathcal{S}} T(s, \pi_t(s), s') \left[r(s, \pi_t(s), s') + V_{t-1}^\pi(s')\right]$$

Infinite Horizons

Assume a stationary transition and reward functions,

- Policy Evaluation:

$$V^{\pi} = \mathbb{E}\left[\sum_{t=0}^{\infty} \gamma^{t} r(S_t, \pi(S_t), S_{t+1}) \middle| \pi, s_0\right]$$

- Value Function:

$$V^{\pi}(s) = \sum_{s' \in \mathcal{S}} T(s, \pi(s), s') \left[r(s, \pi(s), s') + \gamma V^{\pi}(s')\right]$$

- Directly solvable: $O(|\mathcal{S}|^3)$

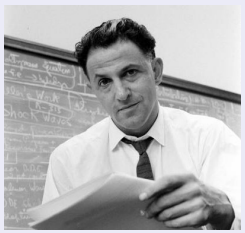
$$\begin{aligned} \mathbf{V}^{\pi} &= r_{\pi} + \gamma \mathbf{T}_{\pi} \mathbf{V}^{\pi} \\ (I - \gamma \mathbf{T}_{\pi}) \mathbf{V}^{\pi} &= \mathbf{r}_{\pi} \\ \mathbf{V}^{\pi} &= (I - \gamma \mathbf{T}_{\pi})^{-1} \mathbf{r}_{\pi} \end{aligned}$$

Bellman's Optimality

Our problem is,

$$\pi^* = \underset{\pi}{\operatorname{argmax}} \left\{ \mathbb{E}\left[\sum_{t=0}^{\infty} \gamma^t r(S_t, \pi(S_t), S_t) \middle| \pi, s_0 \right] \right\}$$

Bellman's Principle of Optimality (?)



"An optimal policy has the property that whatever the initial state and initial decision are, the remaining decisions must constitute an optimal policy with regard to the state resulting from the first decision."

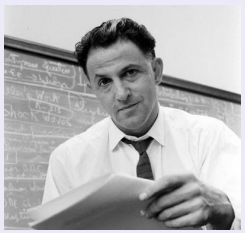
Richard Bellman

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The Value Function (?)



Richard Bellman

$$V_t^*(s) = \max_{a \in \mathcal{A}} \left\{ \sum_{s' \in \mathcal{S}} T(s, a, s') \left[r(s, a, s') + \gamma V_{t-1}^*(s') \right] \right\}$$

Dynamic Programming, Linear Programs, Value Iteration, Policy Iteration, etc...

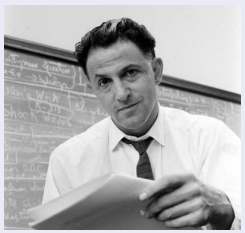
- Complexity: P-Complete for discrete state and action spaces

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The Value Function (?)



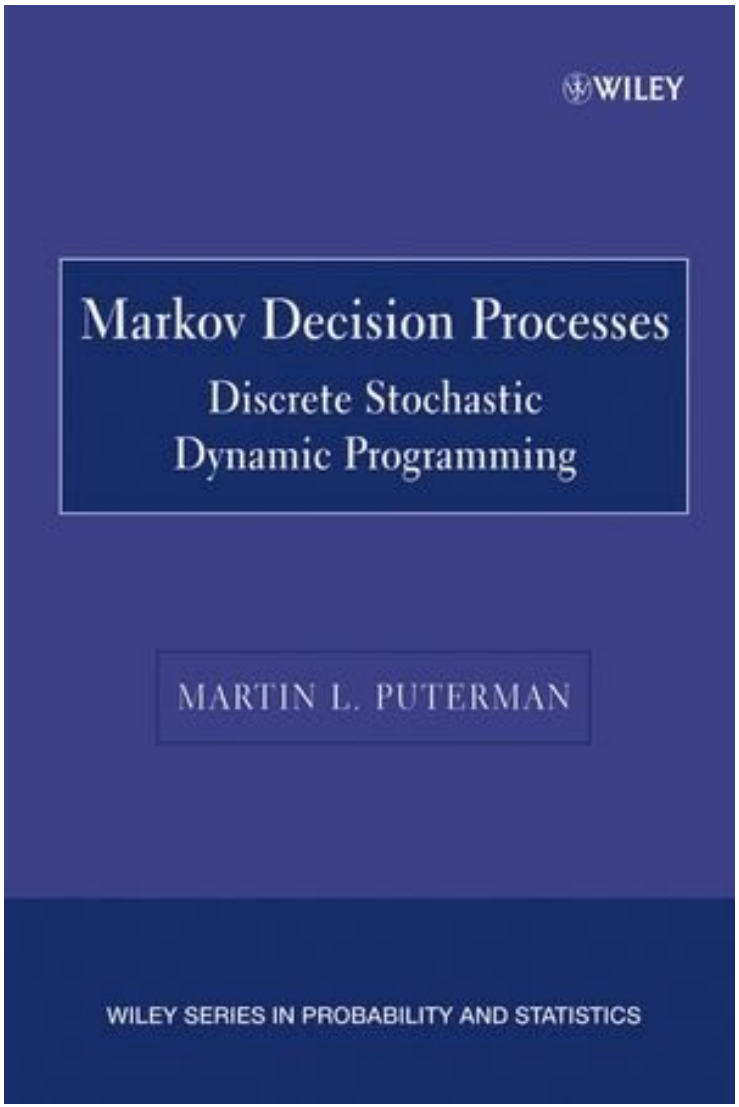
Richard Bellman

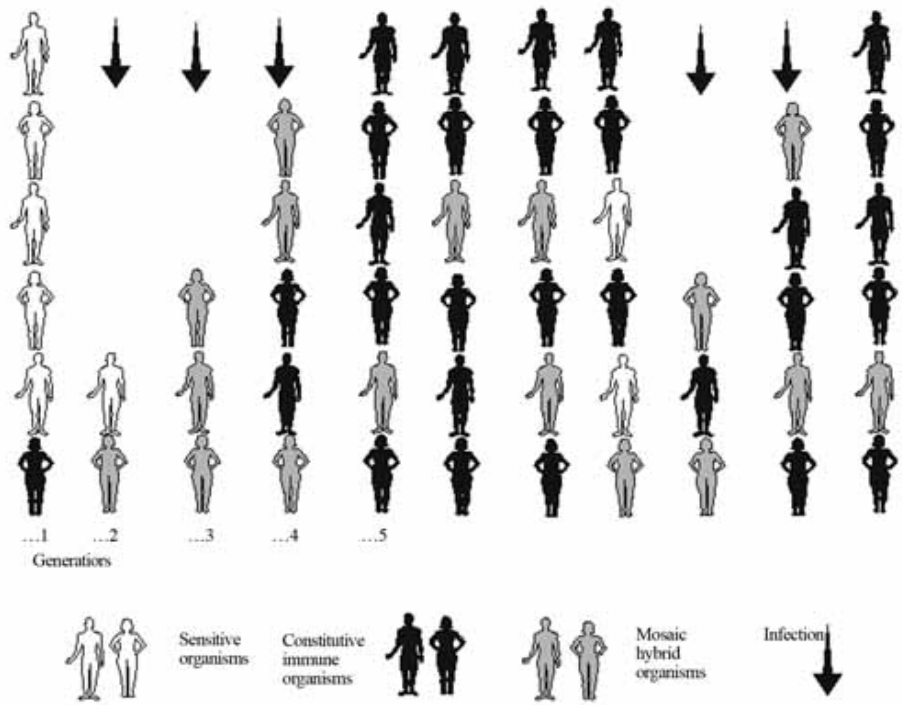
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Dynamic Programming, Linear Programs, Value Iteration, Policy Iteration, etc...

- Complexity: P-Complete for discrete state and action spaces
- Very successful in board games (small state/action space)

More on MDPs


WILEY
Markov Decision Processes
Discrete Stochastic
Dynamic Programming
MARTIN L. PUTERMAN
WILEY SERIES IN PROBABILITY AND STATISTICS


...1
...2
...3
...4
...5
Generations
Sensitive organisms
Constitutive immune organisms
Mosaic hybrid organisms
Infection

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Uncertainty in MDPs

Two types of uncertainty,

1. **Aleatory Variability**: inherent stochasticity or unmodelled variables
   - $T(s, a, s')$ is usually an stochastic representation of complex deterministic dynamics
   - $r(s, a, s')$ may be stochastic also in some applications

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   - **Approximate Dynamic Programming** (ADP): $V^*(s)$ is unknown, because $\mathcal{S}$ or $\mathcal{A}$ are huge

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   - **Partially Observable MDPs** (POMDP): S_t is unknown, because the state is hidden
   - **Reinforcement Learning** (RL): $T(s, a, s')$ is an unknown stochastic function, because the model is not given

A taste of ADP: Approximate Policy Evaluation

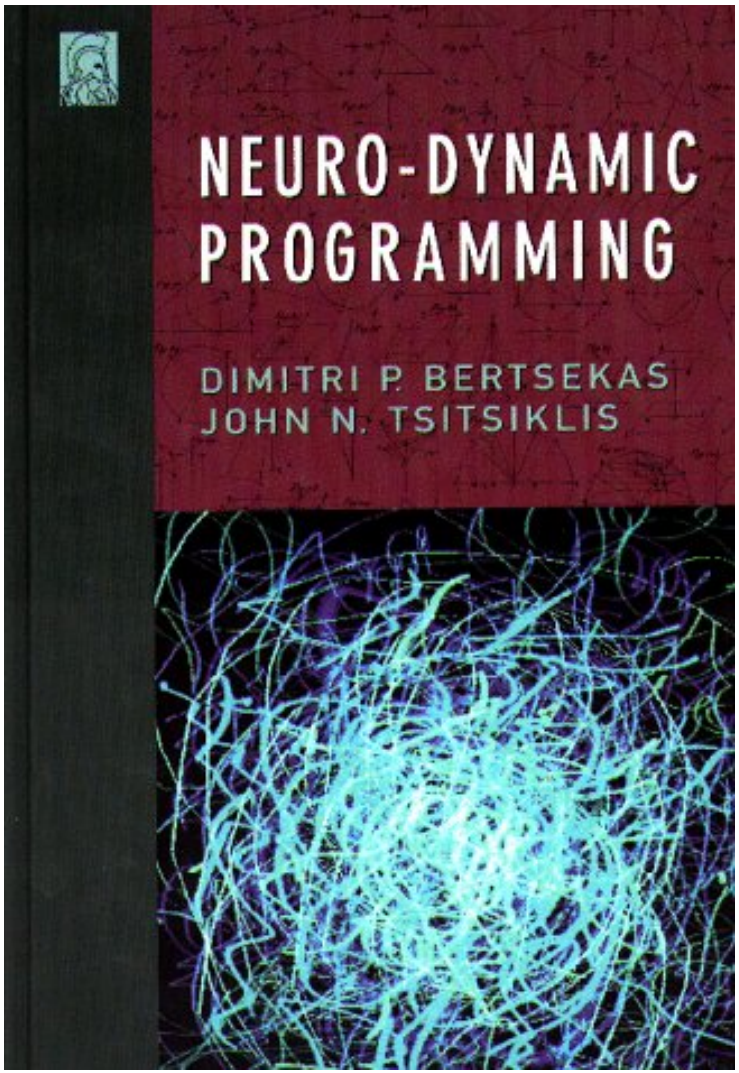
- Approximate Value Function
  $\hat{V}^{\pi}(s;\omega)$: function approximator with parameters ω
- For example a linear combination of basis functions
  $\hat{V}^{\pi}(s;\omega) = \sum_{j=1}^{k} \phi_j(s)\omega_i$
- Apply Least Squares:

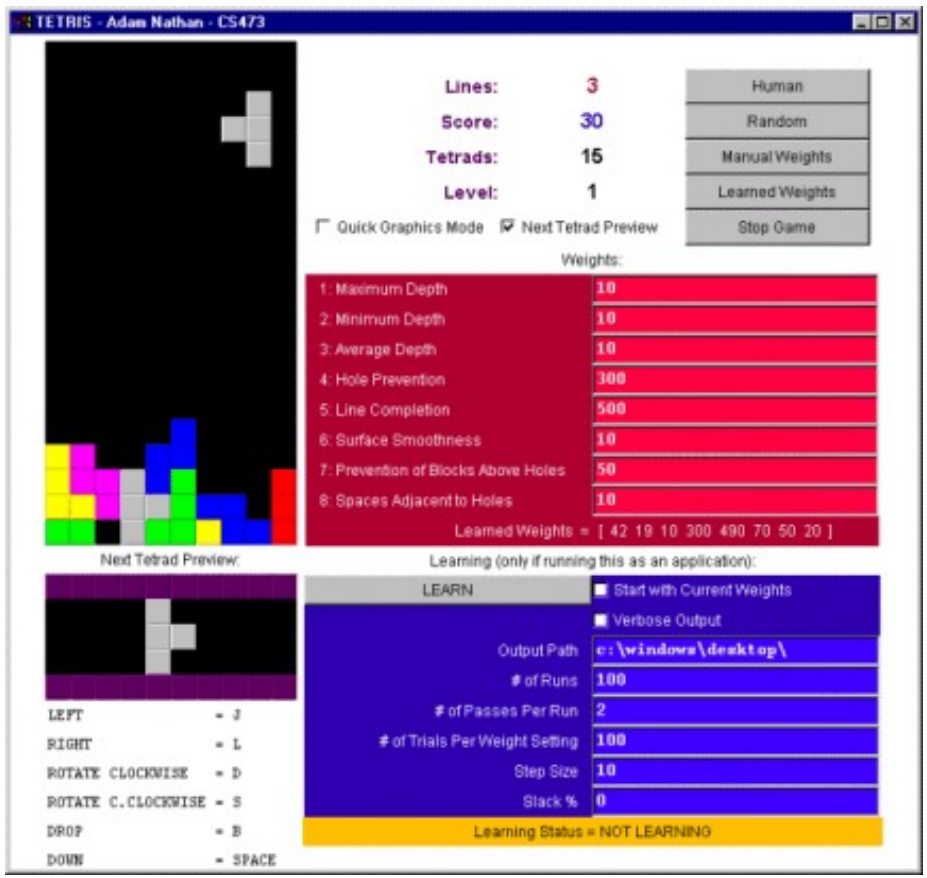
$$\begin{aligned} \hat{\mathbf{V}}^{\pi} &\approx r_{\pi} + \gamma \mathbf{T}_{\pi} \hat{\mathbf{V}}^{\pi} \\ \mathbf{\Phi}\omega^{\pi} &\approx r_{\pi} + \gamma \mathbf{T}_{\pi} \mathbf{\Phi}\omega^{\pi} \\ (\mathbf{\Phi} - \gamma \mathbf{T}_{\pi} \mathbf{\Phi})\omega^{\pi} &\approx r_{\pi} \end{aligned}$$

$$\underset{\omega^{\pi}}{\operatorname{argmin}} \left\| (\mathbf{\Phi} - \gamma \mathbf{T}_{\pi} \mathbf{\Phi})\omega^{\pi} - r_{\pi} \right\|_2 \tag{1}$$

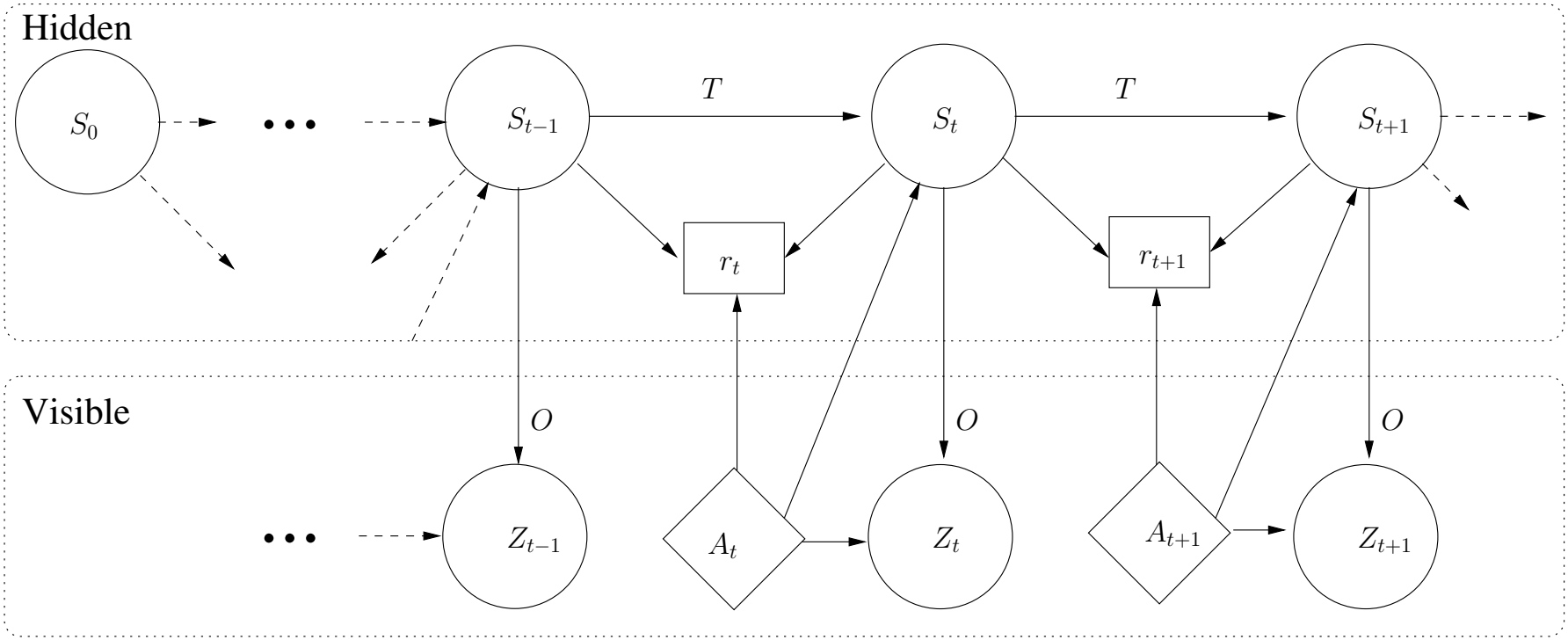
- The parameters can be learned by samples, simulations or analytically

Approximate Dynamic Programming





Partially Observable MDP



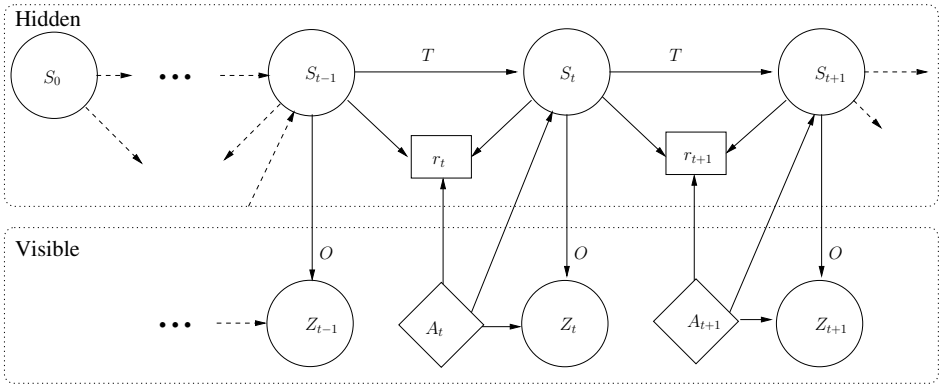
POMDP

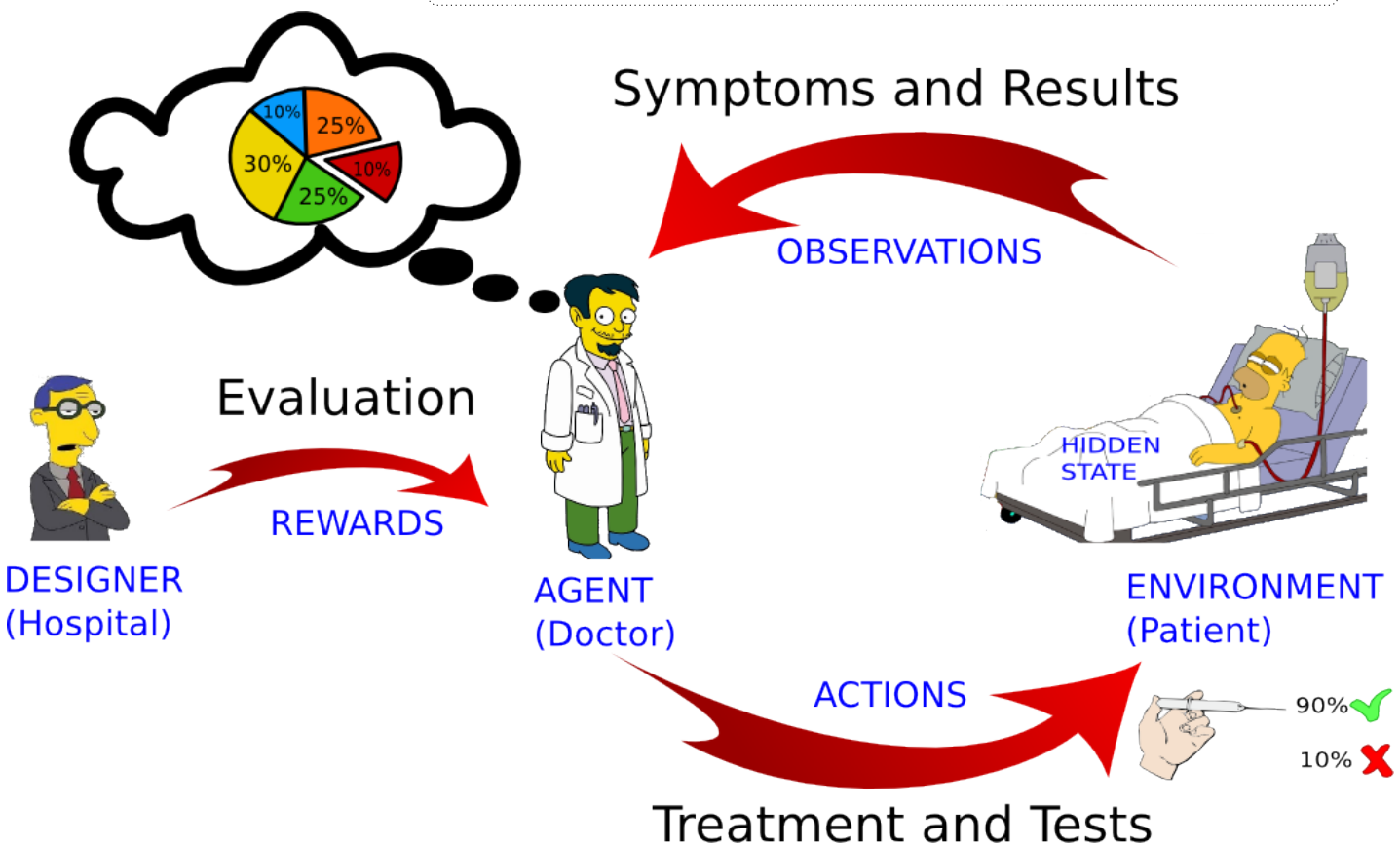
States are hidden $\longrightarrow$ Observations

Observation Function: $O(s', a, z) = Pr(z|s', a)$

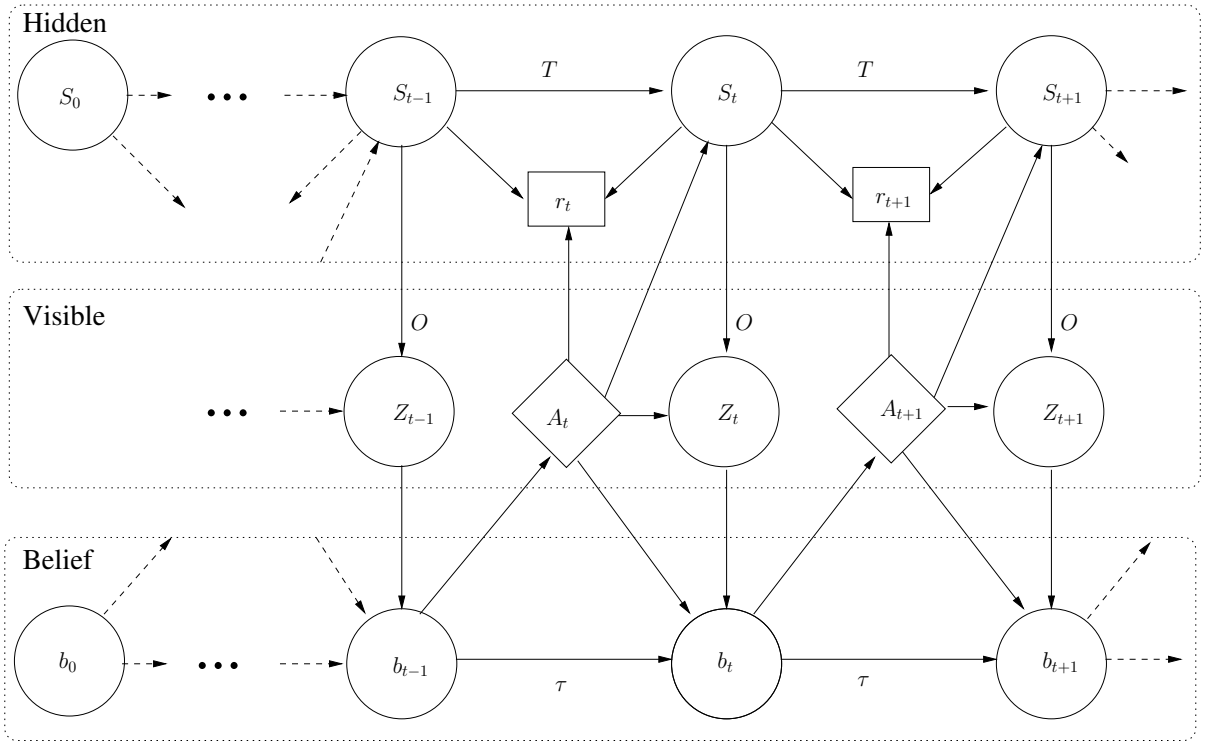
$\langle \mathcal{S}, \mathcal{A}, \mathcal{Z}, T, O, r \rangle$

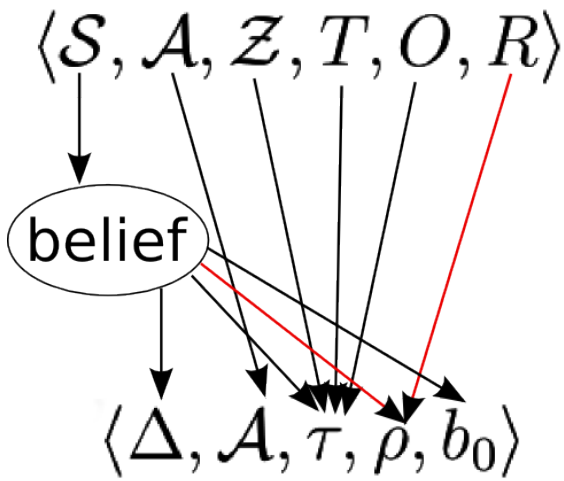
Agent's Belief





Belief-MDP





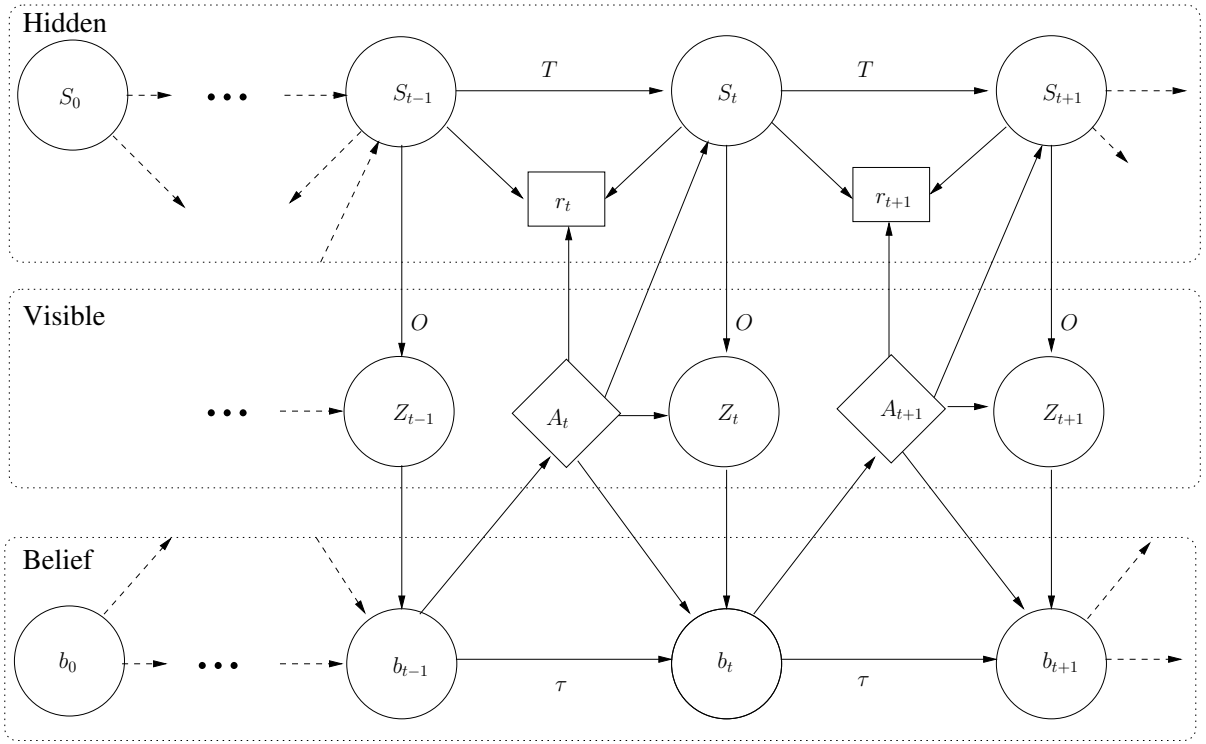
Belief-MDP (?)

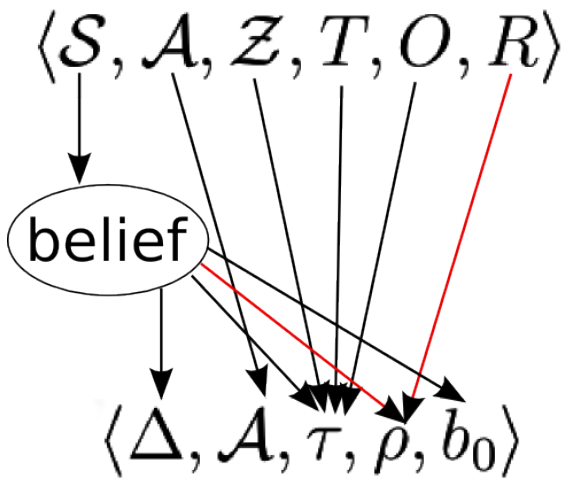


Karl Åström

- **Belief-state**: Probability distribution over the states
- Is a sufficient statistic of history
- Bayes rule + Graphical models = **Belief-MDP**

Belief-MDP

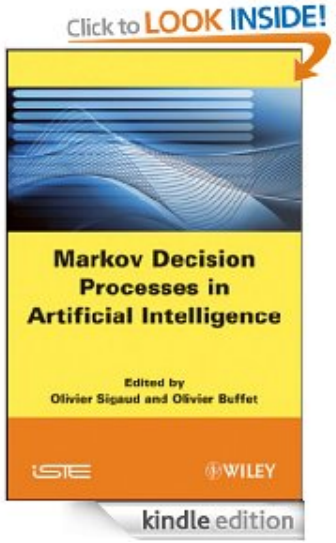


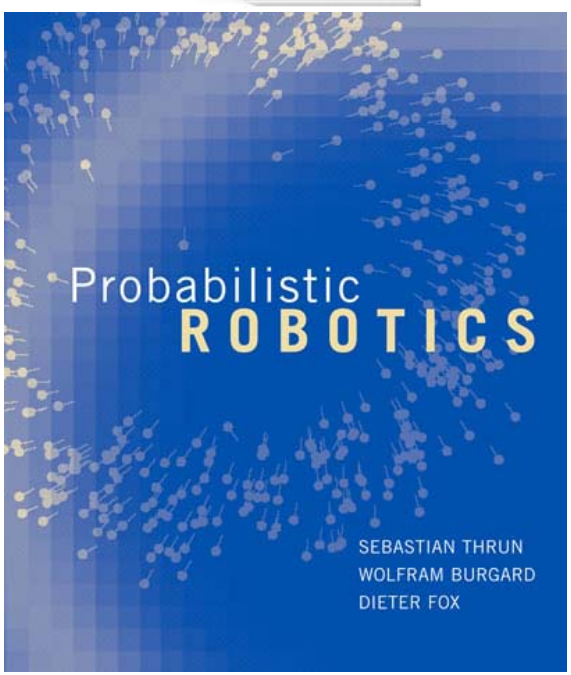


Belief-MDP Bellman Equation

- Belief-MDP tuple: $\langle \Delta, \mathcal{A}, \tau, \rho, b_0 \rangle$
- $V_t^*(\mathbf{b}) = \max_{a \in \mathcal{A}} \left\{ \rho(b,a) + \gamma \int_{\mathbf{b}' \in \Delta} \tau(\mathbf{b}, a, \mathbf{b}') V_{t-1}^*(\mathbf{b}, a) d\mathbf{b}' \right\}$
- $\rho(b,a) = \sum_{s \in \mathcal{S}} b(s) R(s,a)$ (expected reward)

More on POMDPs





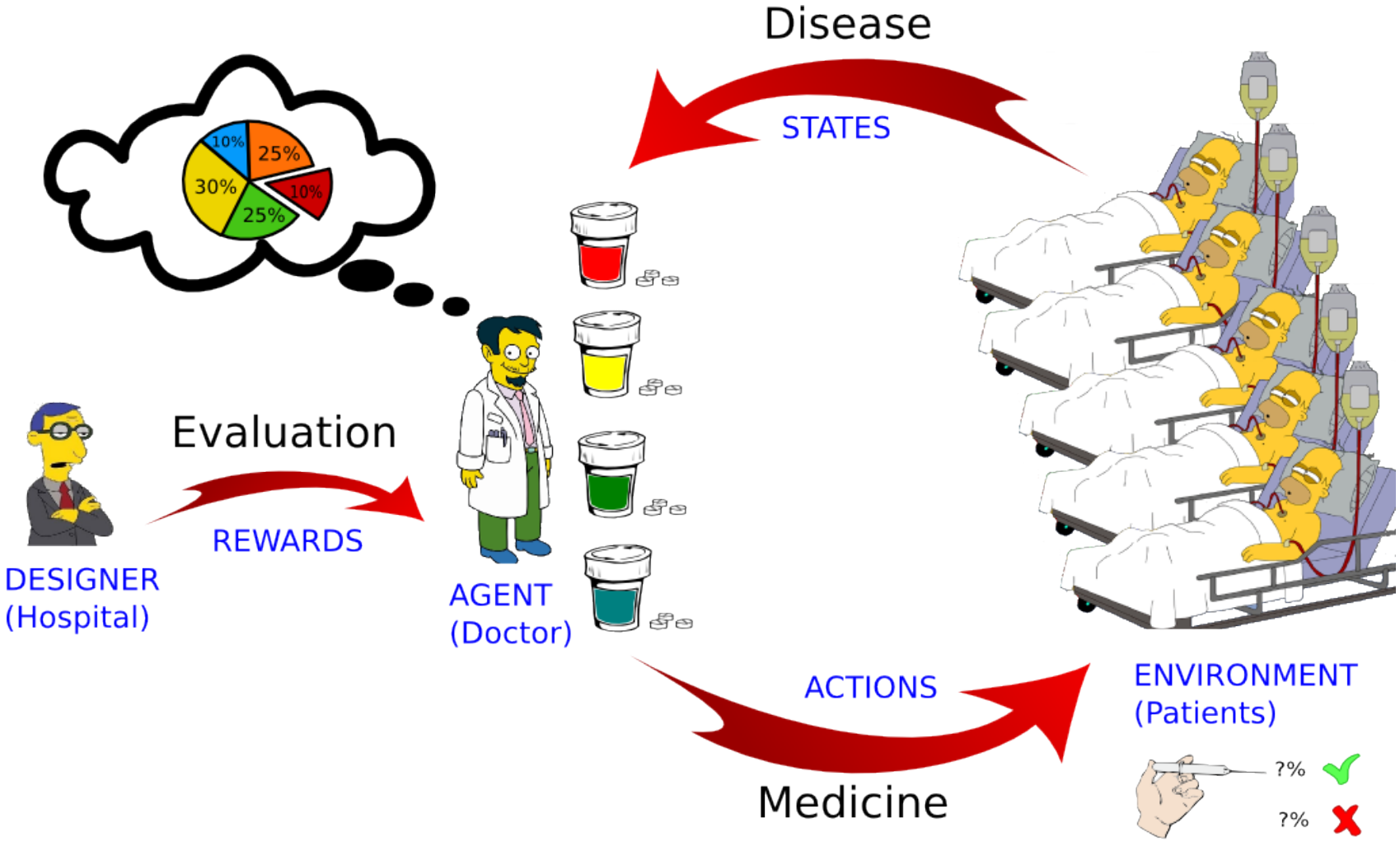




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Clinical Trials — Bandit Problem (?)



- The objective is to maximize the number of **healthy** patients
- But the **effect** of medicine is partially known (must be learned)

Reinforcement Learning

Reinforcement Learning (RL) (?)

- Let $M = \langle \mathcal{S}, \mathcal{A}, T, R, s_0 \rangle$ be an MDP, and π a policy.
- $V_T^\pi(s) = \mathbb{E}_\pi \left[\sum_{t=0}^{\infty} \gamma^t r(s_t, a_t, s_{t+1}) \middle| s_0 = s, T \right]$
- How to find a π that maximizes V if T is unknown?
- Dilemma: to learn or to gain rewards?



Richard Sutton

Reinforcement Learning

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Richard Sutton

Solution Paradigms

- Model-based or Model-free RL
- Value iteration or Policy Iteration
- Intelligent or Random Exploration
- Bayesian or "Frequentist" RL

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Model-based and Model-free RL

Simple Model-based RL (?)

At each step t,

- estimate $\hat{T}$ from observed transitions
- follow an exploitation policy $\hat{\pi} \in \operatorname{argmax}_\pi \{V^\pi_{\hat{T}}(s_t)\}$,
- or follow some exploration policy that improves $\hat{T}$.

Problem: exploration/exploitation dilemma



Leslie Kaelbling

Model-based and Model-free RL

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Leslie Kaelbling

Simple Model-free RL (?)

At each step t, use (s, a, s', r) to

- update a state-action value function:
  $Q_t(s,a) = Q_{t-1}(s,a) + \alpha \left(r + \max_{a' \in \mathcal{A}} \{Q_{t-1}(s', a')\} - Q_{t-1}(s,a) \right),$
- learns the Q-value function, act greedily w.r.t. Q

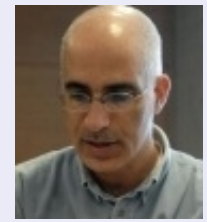
Problem: very slow convergence


Chris Watkins

Optimistic Algorithms

RMAX (?)

At each time step t,



- For each (s, a)

$$Q(s,a) = R(s,a) + \gamma \sum_{s' \in \mathcal{S}} \hat{T}(s,a,s') \max_{a'} Q(s',a') \quad \text{if } n(s,a) \geq m,$$

$$Q(s,a) = \frac{R_{max}}{1-\gamma} \quad \text{otherwise,}$$

- $n(s,a) =$ how many times a (s,a) transition has occurred
- use (s,a,s',r) to update $\hat{T}(s,a,s') = \frac{n(s,a,s')}{n(s,a)}$
- Solve a fixed point solution and use that policy for a while

From RL to Bayesian (Model-based) RL

Bayesian Model-based RL (?)

- T is a random variable $\sim Pr(T|\mathbf{b})$
- The parameters $\mathbf{b}$ are the **belief over the models**
- $\mathbf{b}$ is updated using the Bayes rule
- Optimal Bayesian policy, $\underset{\pi}{\operatorname{argmax}}\, \mathbb{V}^{\pi}(s_t, \mathbf{b})$

From RL to Bayesian (Model-based) RL

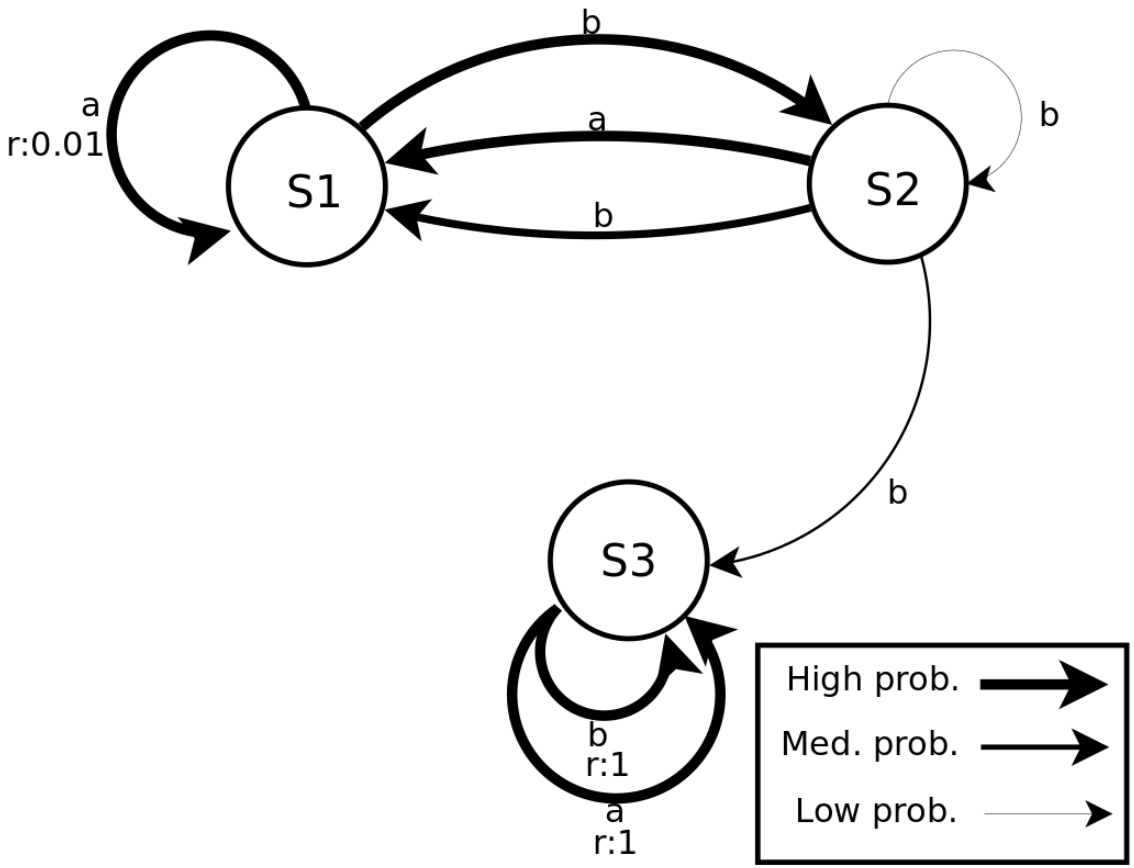
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- Optimal Bayesian policy, $\underset{\pi}{\operatorname{argmax}}\, \mathbb{V}^{\pi}(s_t, \mathbf{b})$
- No **exploration/exploitation** dilemma

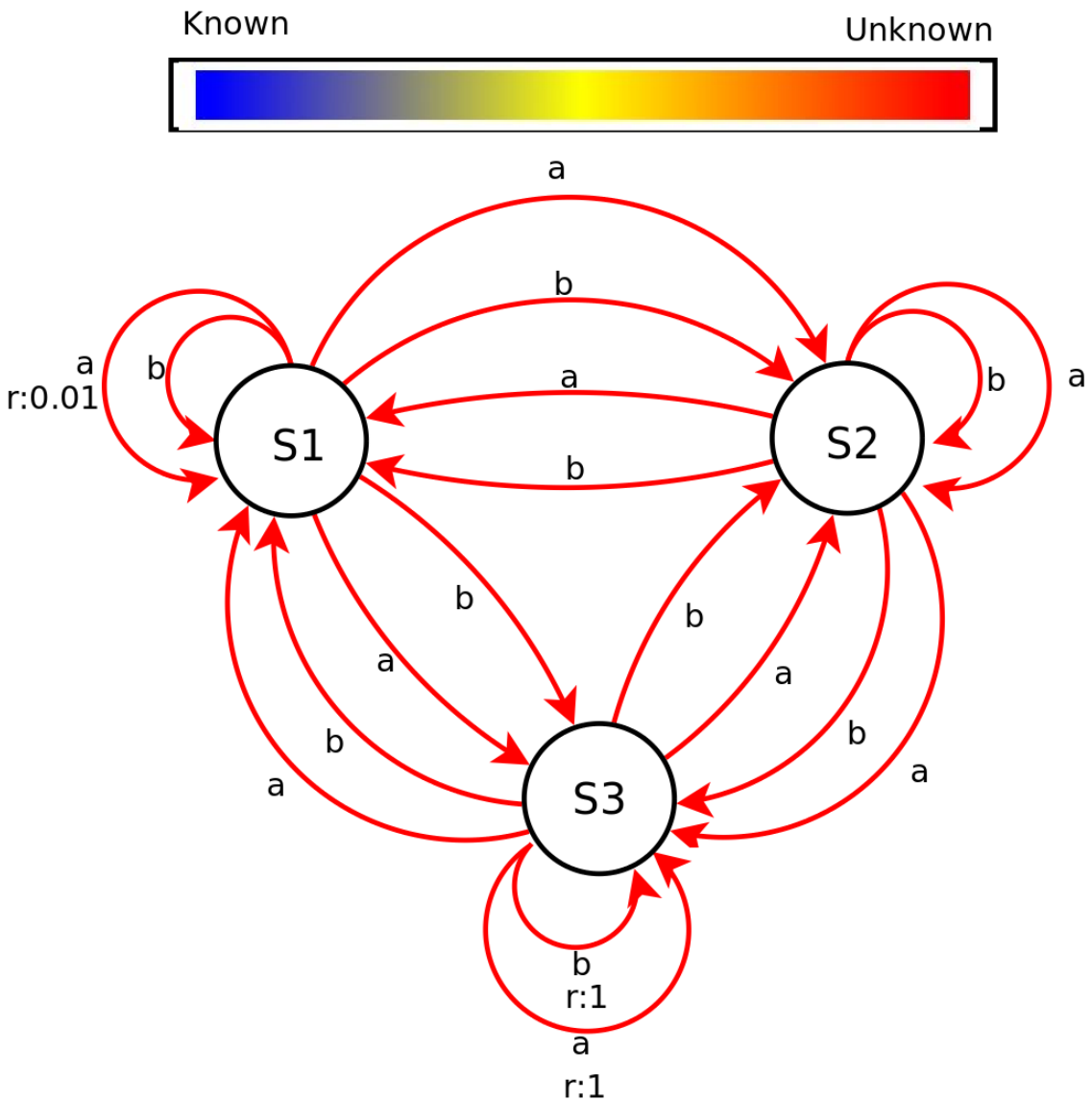


Michael O. Duff

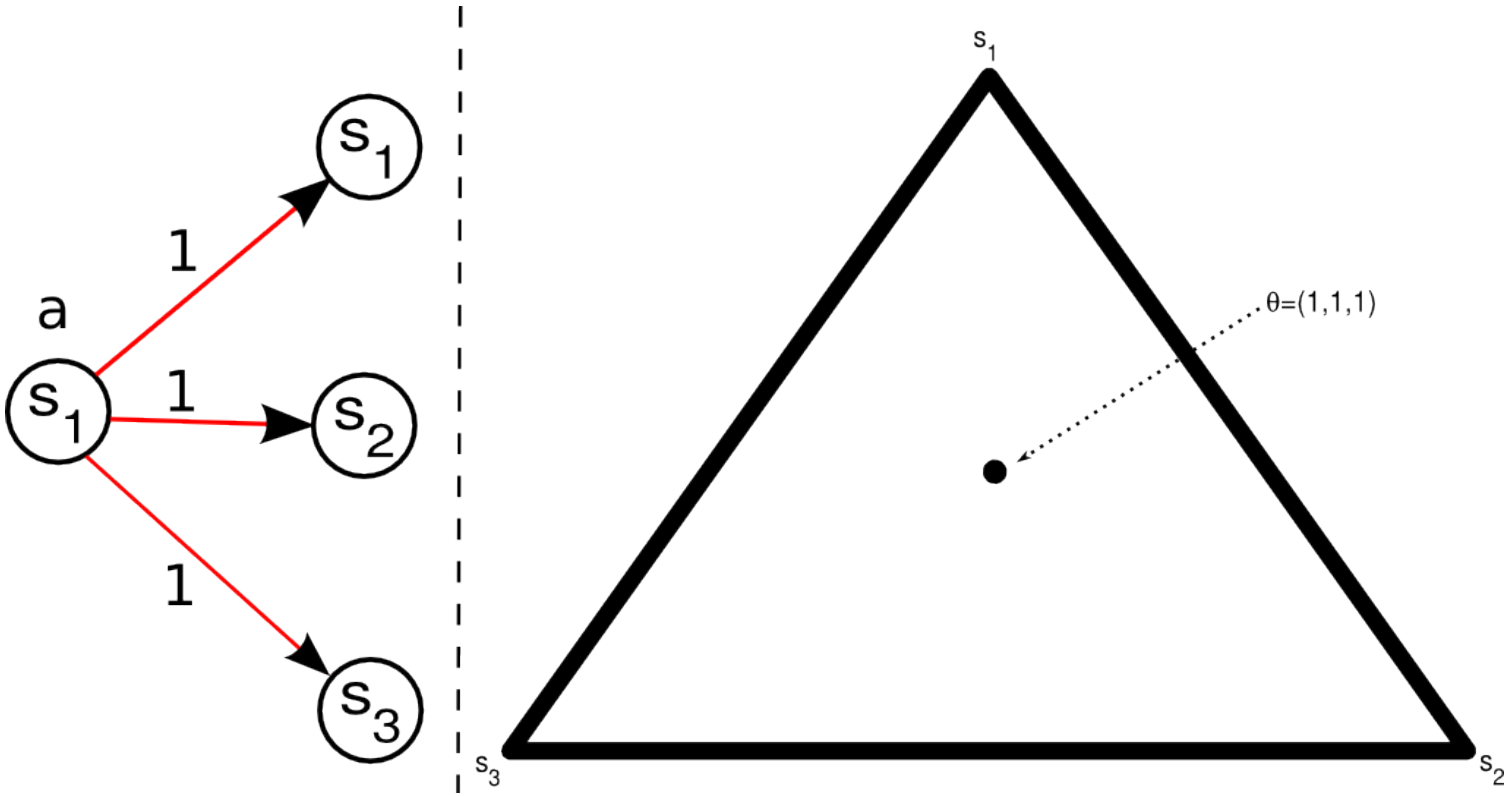
Bayesian Modelling Example



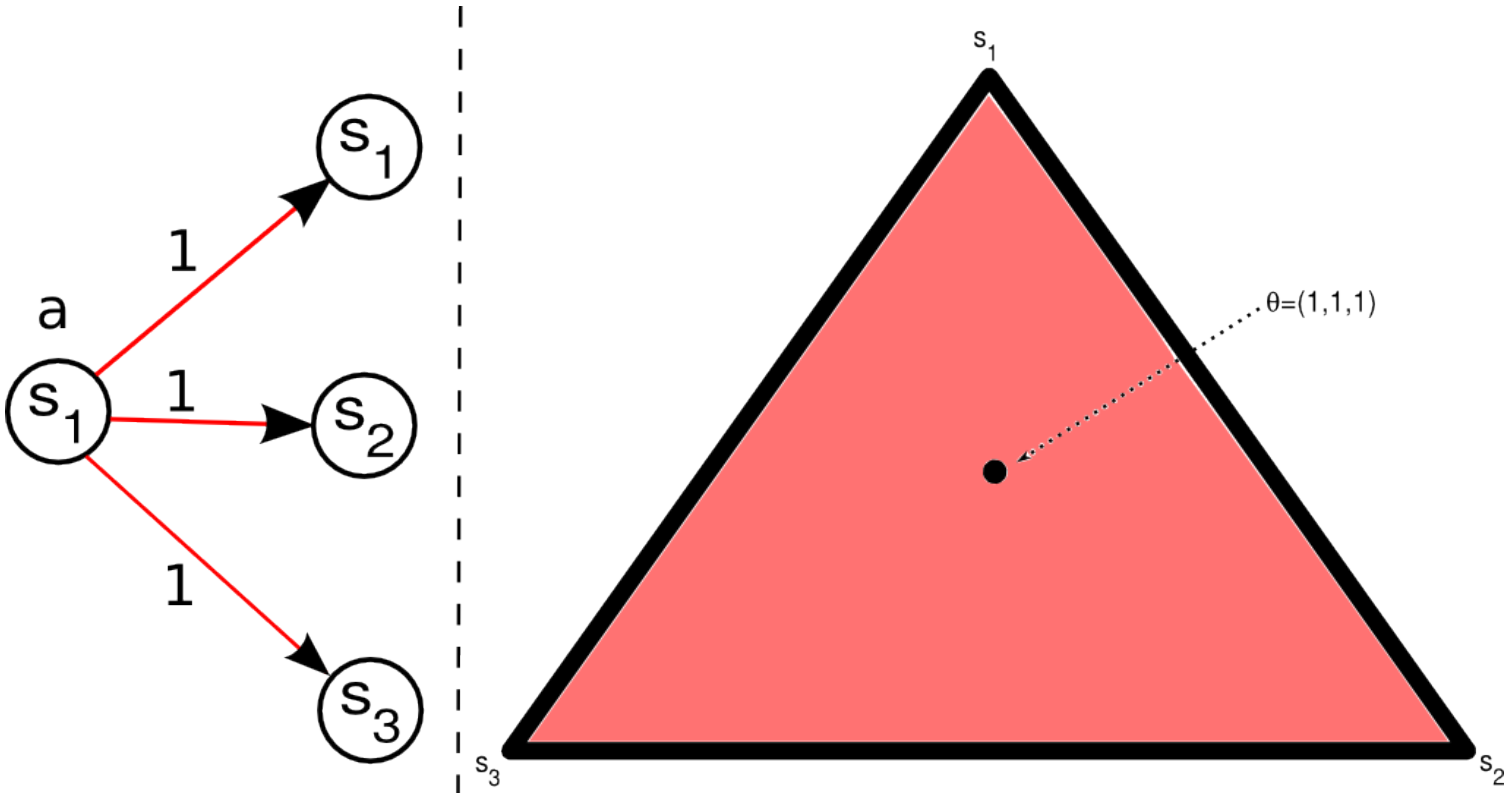
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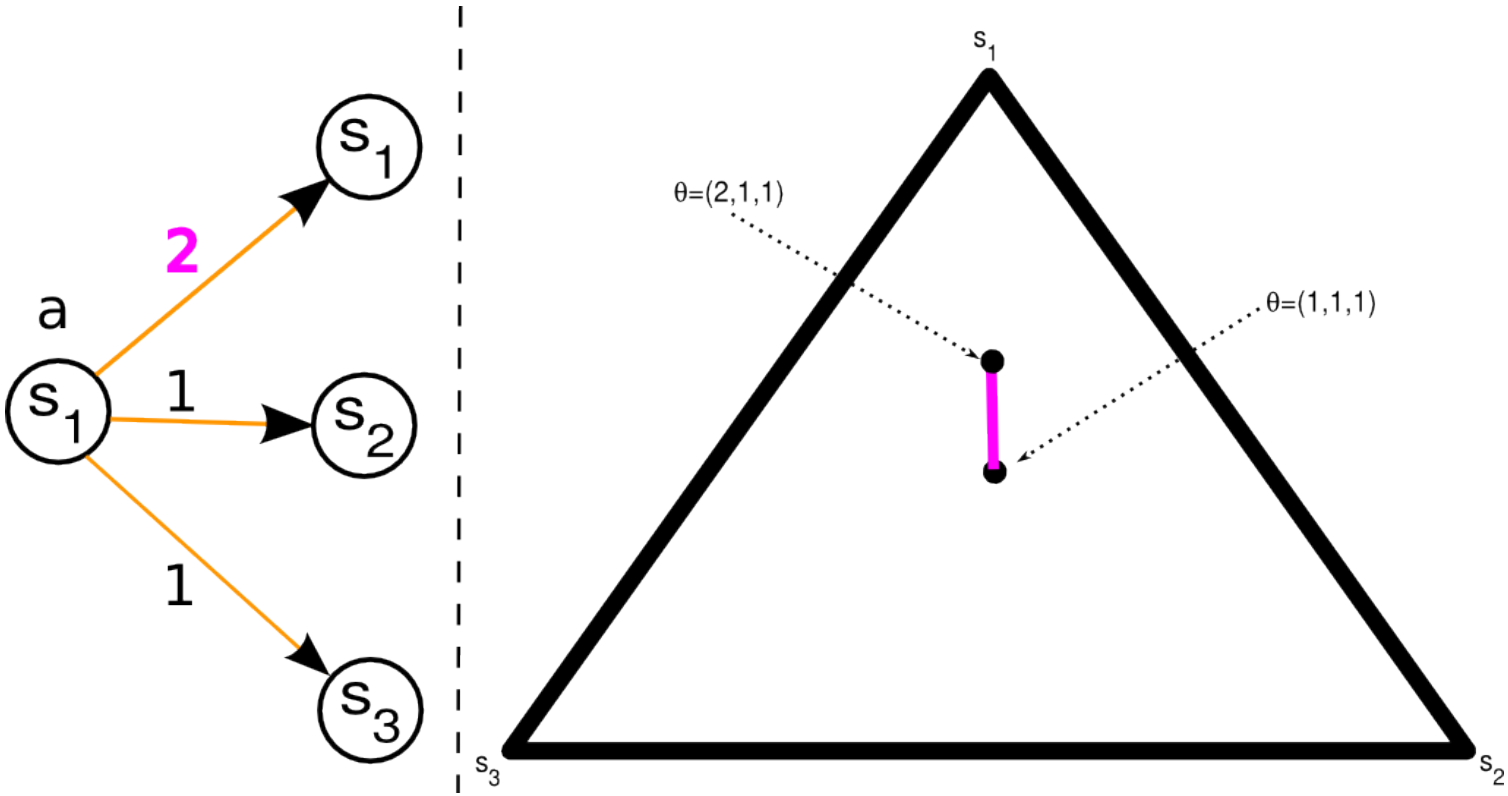
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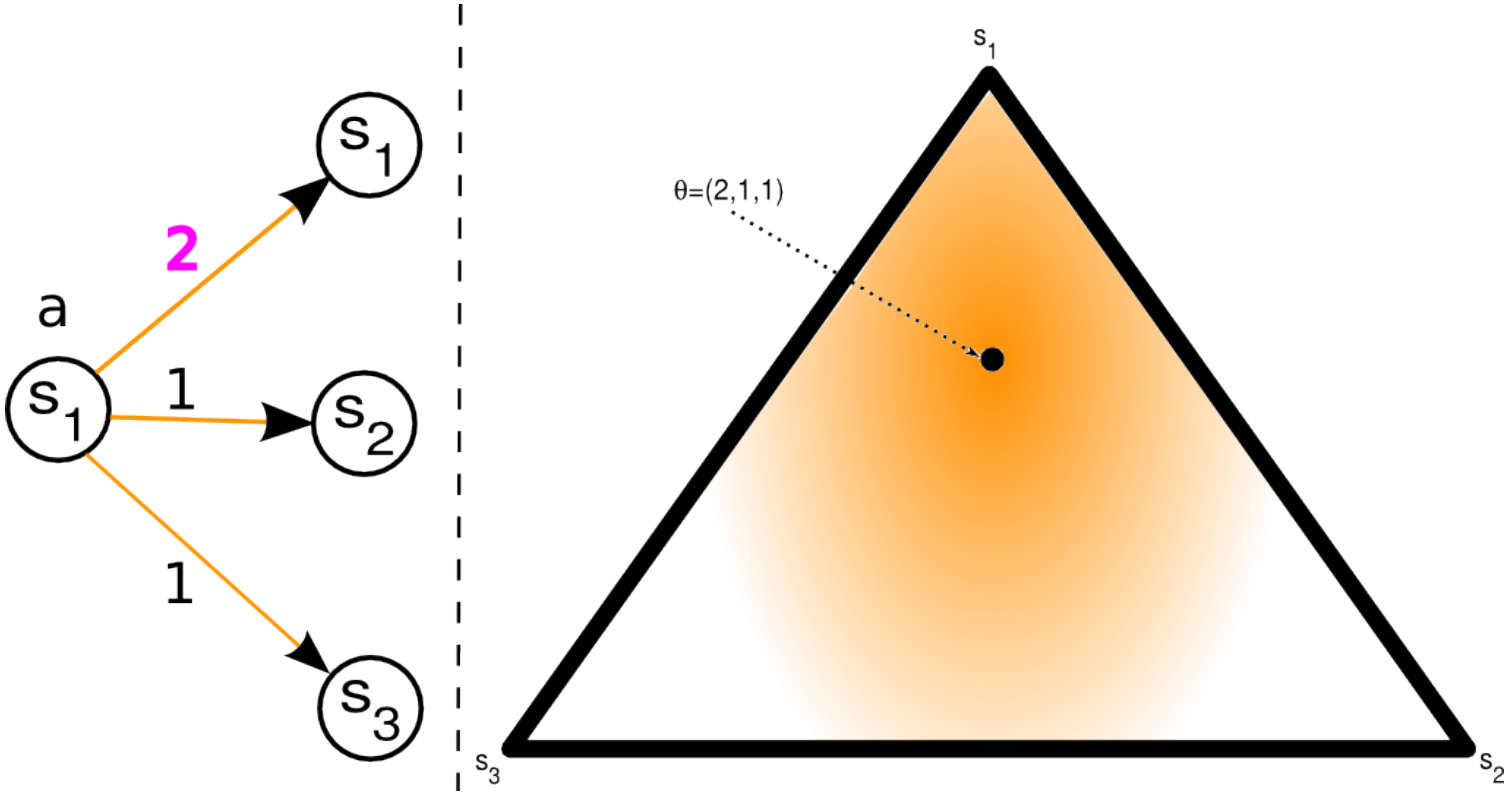
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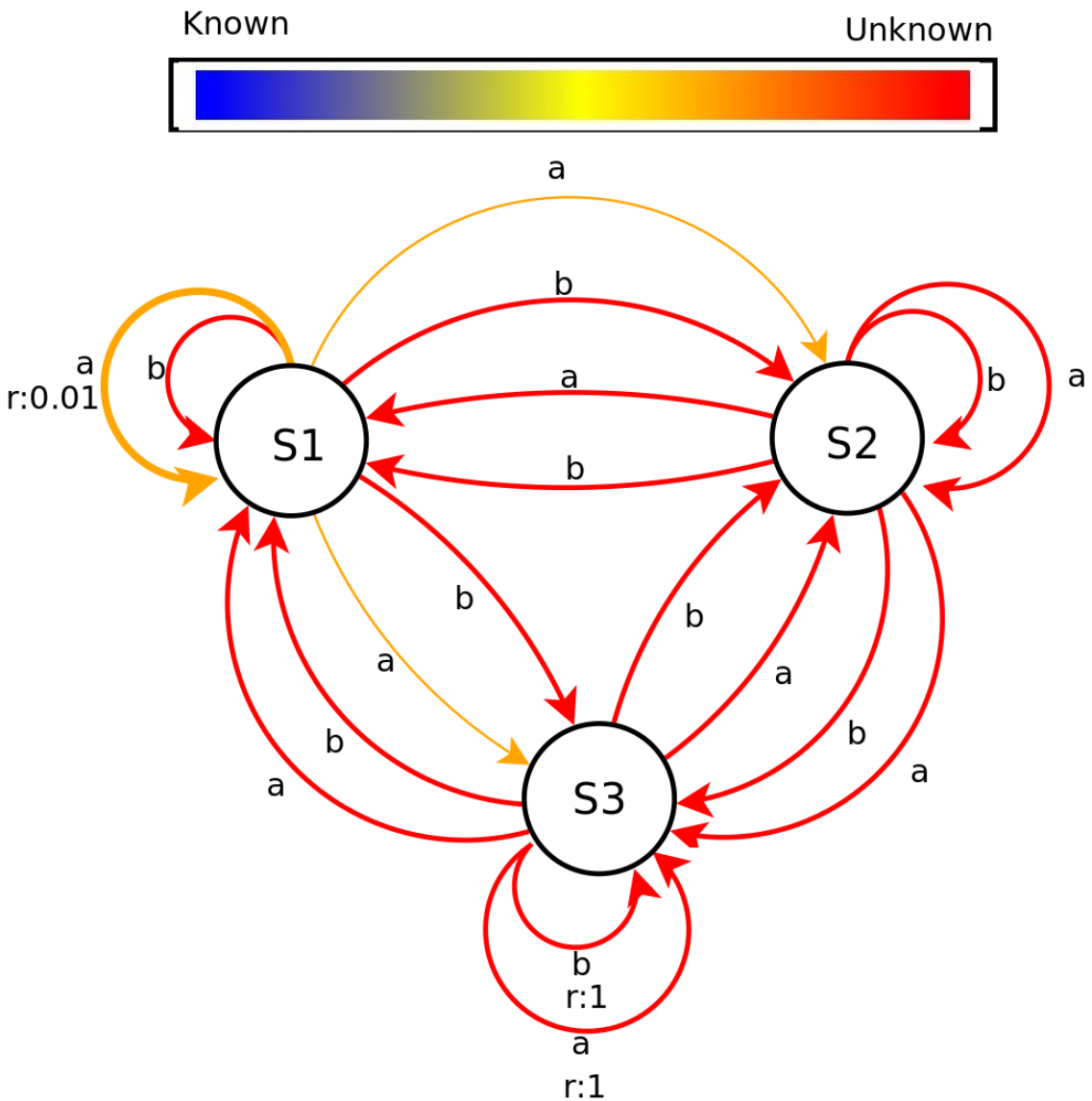
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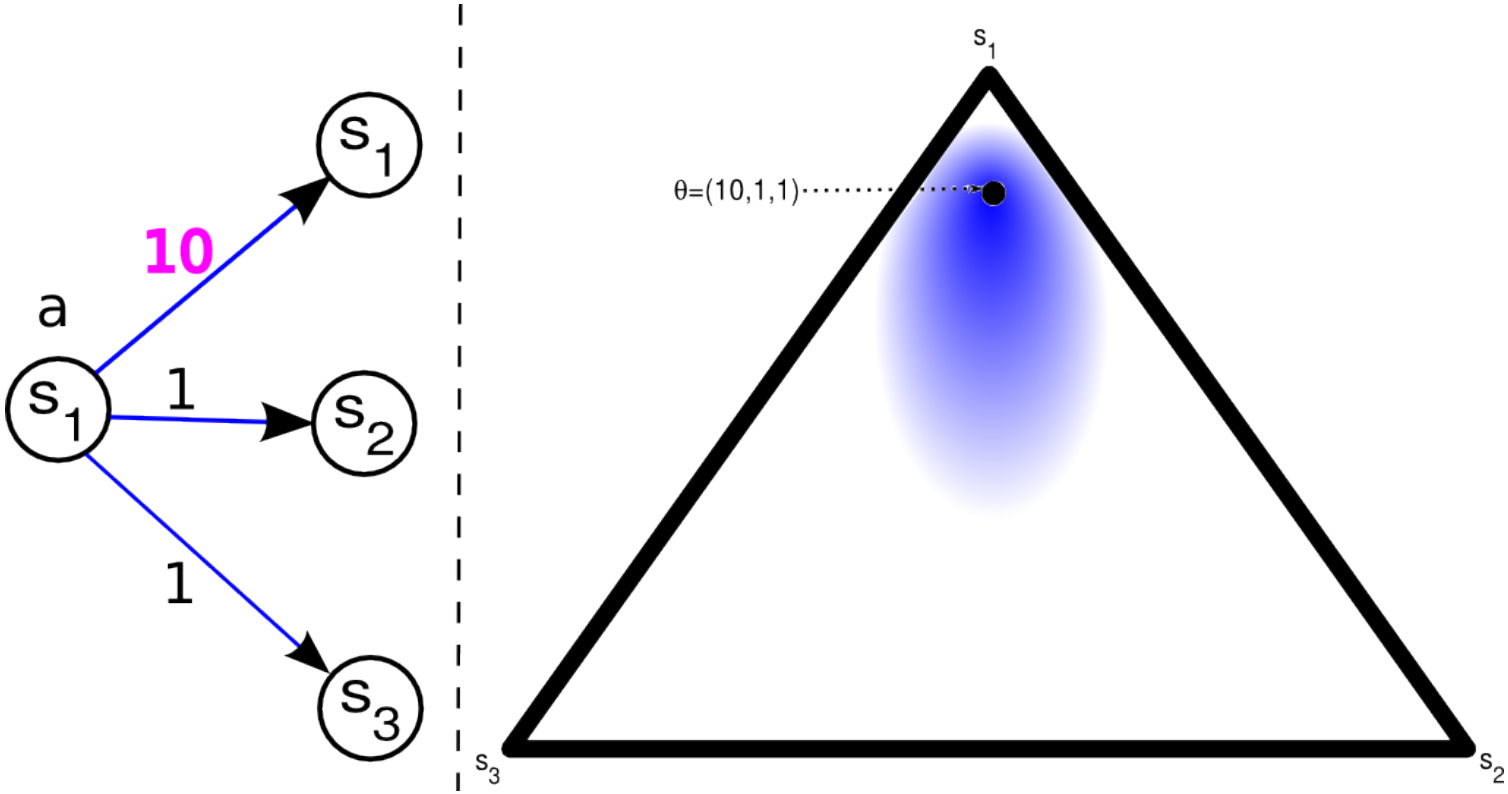
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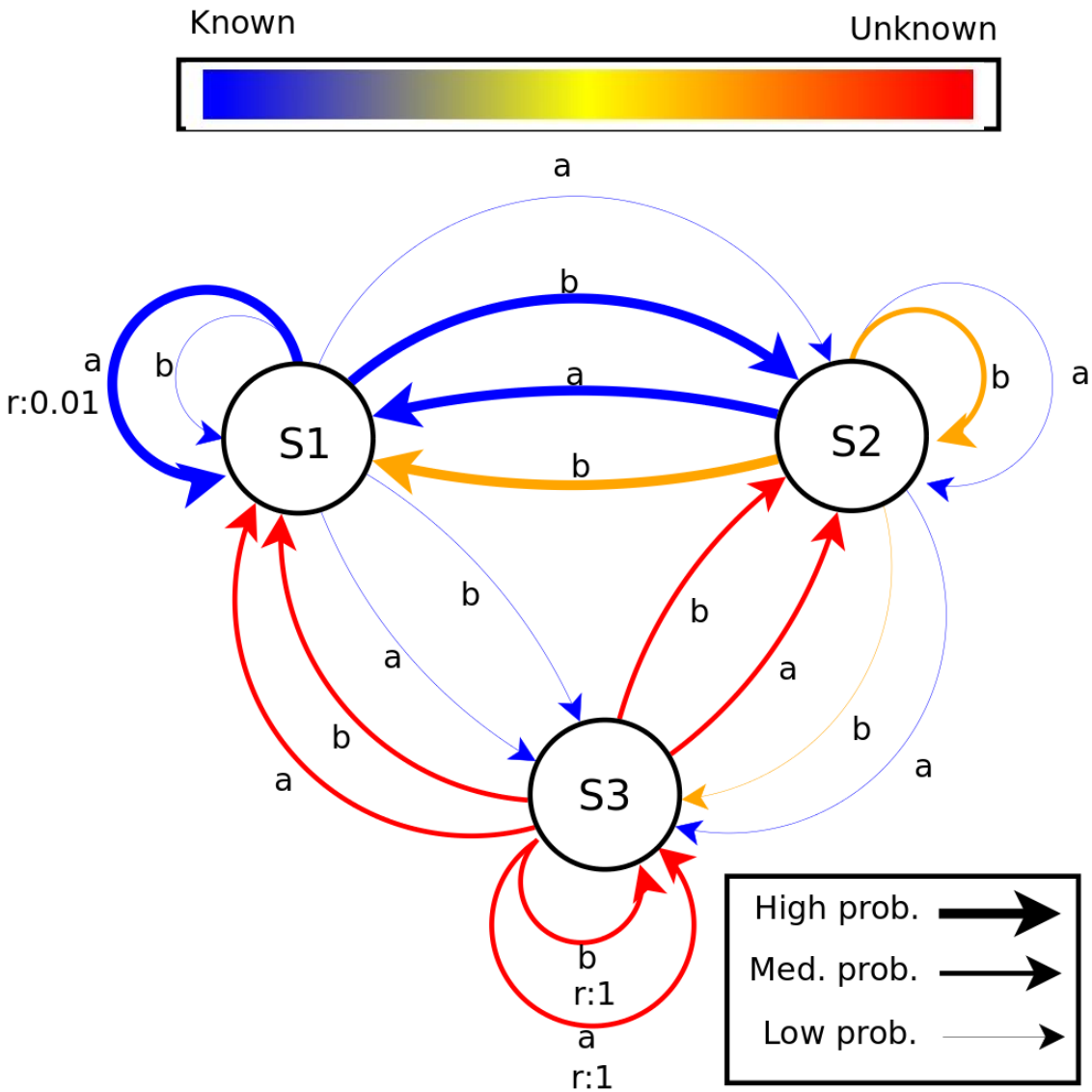
Bayesian Modelling Example



Bayesian Modelling Example



Bayesian Modelling Example



Approximate Bayesian RL

Computing $\mathbb{V}$ is intractable

Approximate Bayesian RL

BEETLE: A Belief-lookahead Approach (?)

- Put T in the state (POMDP)
- Approximately solve an infinite-dimensional belief-MDP
- Elegant and sound
- Slow and inefficient



Pascal Poupart

Approximate Bayesian RL

BEETLE: A Belief-lookahead Approach (?)

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- Approximately solve an infinite-dimensional belief-MDP
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Pascal Poupart

Optimistic (Myopic) Approaches



(?)

BOSS: Best Of the Sampling Set

- Is optimistic about the transition function
- Use sampling (slow and unreliable)
- Several parameters

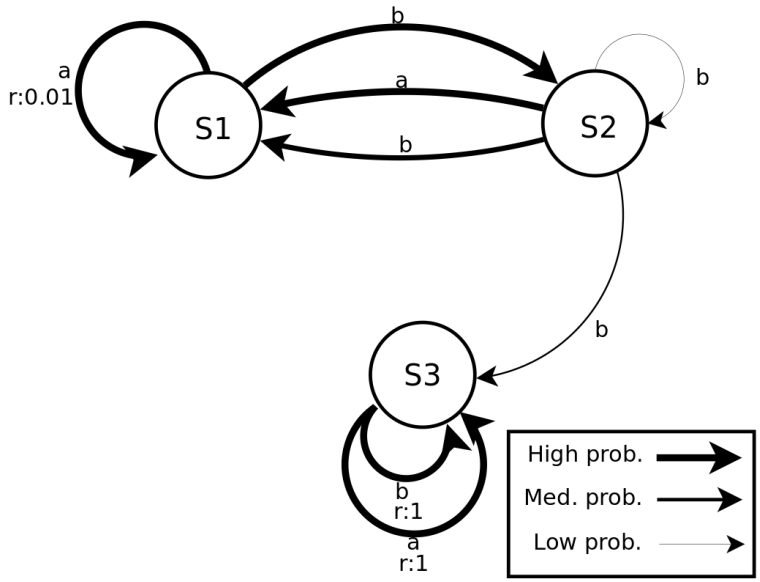


(?)

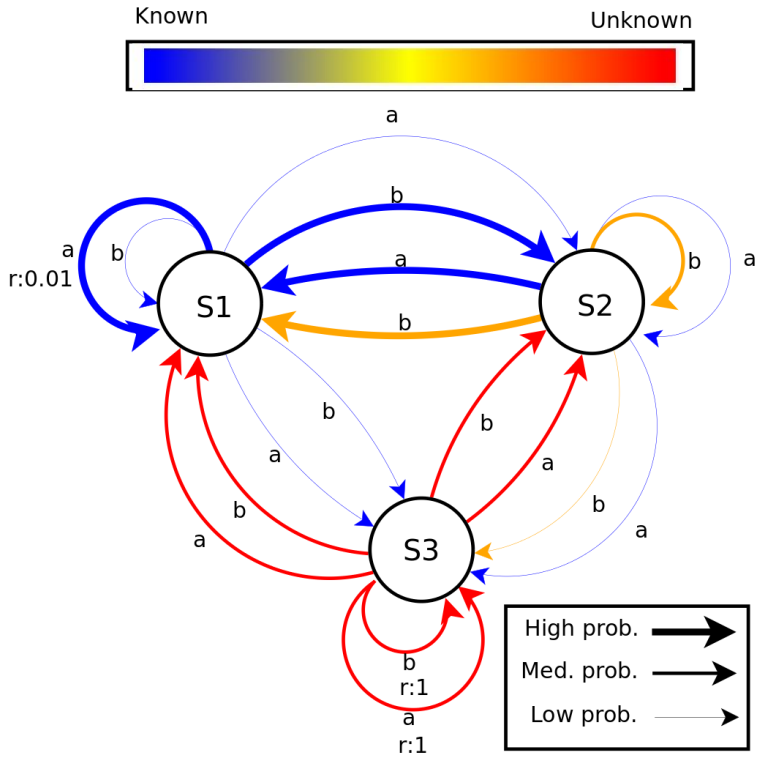
BEB: Bayesian Exploration Bonus

- Simple and fast
- Only one parameter
- The optimism is unbounded (rewards)

A Simple Myopic Algorithm: EXPLOIT



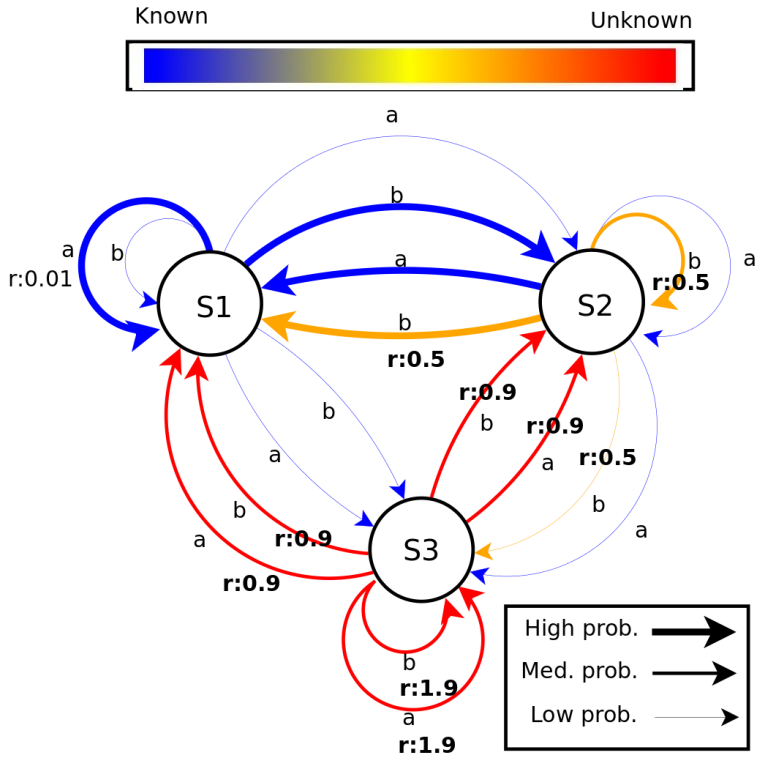
A Simple Myopic Algorithm: EXPLOIT



At each step t:

1. Update the belief $\mathbf{b}' = Bayes(\mathbf{b}, s, a, s')$
2. Compute the expected model $\hat{T} = \mathbb{E}[T|\mathbf{b}]$
3. Compute a policy in $\text{argmax}_\pi \hat{V}^\pi_{\hat{T}}(s)$

A Simple Optimistic Algorithm: BEB



At each step t:

1. Update the belief $\mathbf{b}' = Bayes(\mathbf{b}, s, a, s')$
2. Compute the expected model $\hat{T} = \mathbb{E}[T|\mathbf{b}]$
3. Add a exploration bonus to R $\frac{\beta}{1+\#_{s,a}}$
4. Compute a policy in $\text{argmax}_\pi \hat{V}^\pi_{\hat{T}}(s)$

Why Optimistic Algorithms are Interesting?

PAC-MDP (?)



$\mathcal{A}$ is PAC-MDP if, with probability $1-\delta$, $V_T^{\mathcal{A}}$ is ϵ-close to V_T^* for all but a **polynomial** number of steps.

- Extends PAC analysis (**?**) to MDPs
- Formally, $\#ns \leq poly(|S|, |A|, \frac{1}{\gamma}, \frac{1}{\delta}, \frac{1}{\epsilon})$
- R-MAX and BOSS are PAC-MDP

PAC-BAMDP (?)



$\mathcal{A}$ is PAC-BAMDP if, with high probability,

$$\mathbb{V}^*(s, \mathbf{b}) - \mathbb{V}^{\mathcal{A}}(s, \mathbf{b}) < \epsilon$$

for all but a **bounded** number of steps.

- Also called near-Bayesian property
- BEB was the first PAC-BAMDP algorithm

Contents

1 Markov Decision Processes

2 Epistemic Uncertainty in MDPs

3 Reinforcement Learning

4 My Contribution

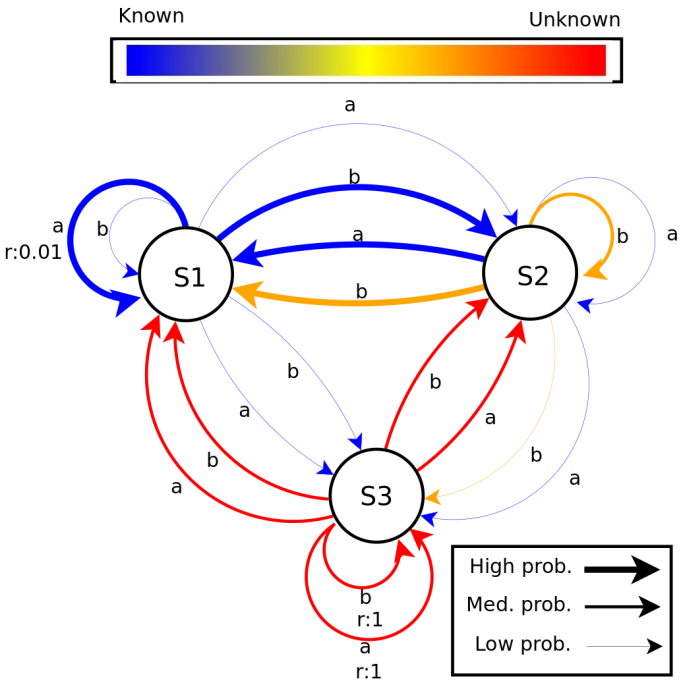
My PhD Thesis

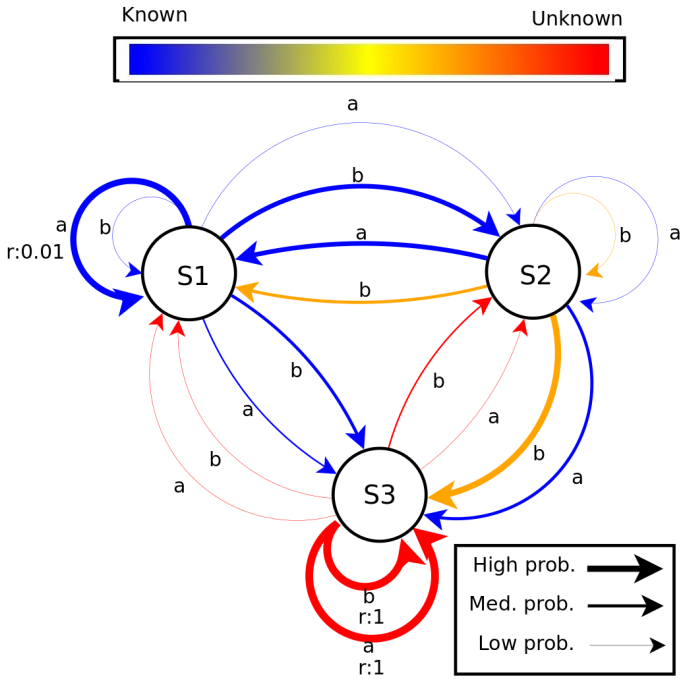
Near-Optimal Algorithms for Sequential Information-Gathering Decision Problems

- BOLT: an Optimistic Bayesian RL algorithm
- ρ-POMDPs: POMDPs for Explicit Information Gathering
- ρ-BRL: Bayesian RL for Optimal Exploration

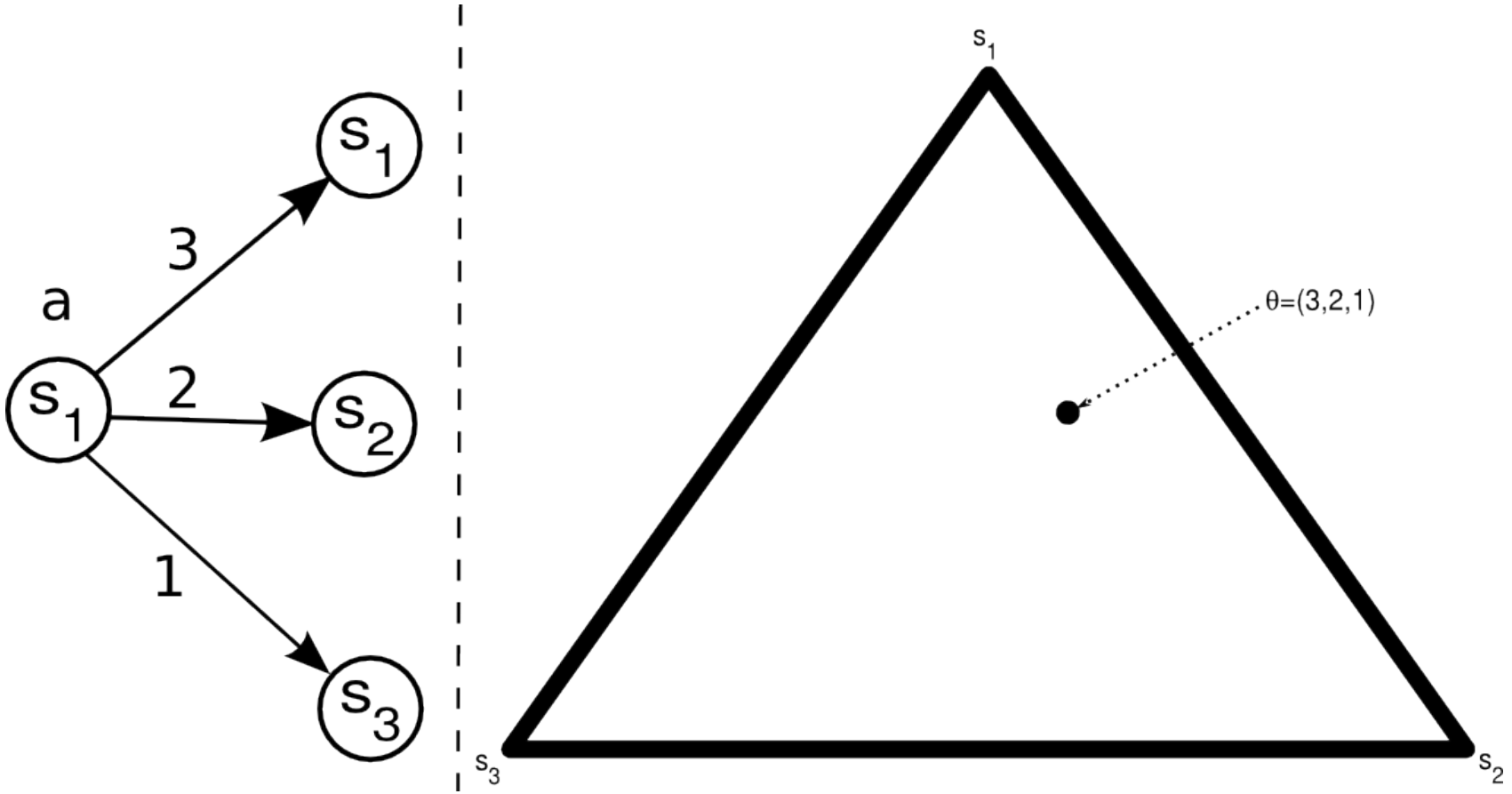
BOLT: Bayesian Optimistic Local Transitions

- **Idea**: be fast and optimistic about the transitions
- **(ICML'12) Bayesian Optimistic Local Transitions**

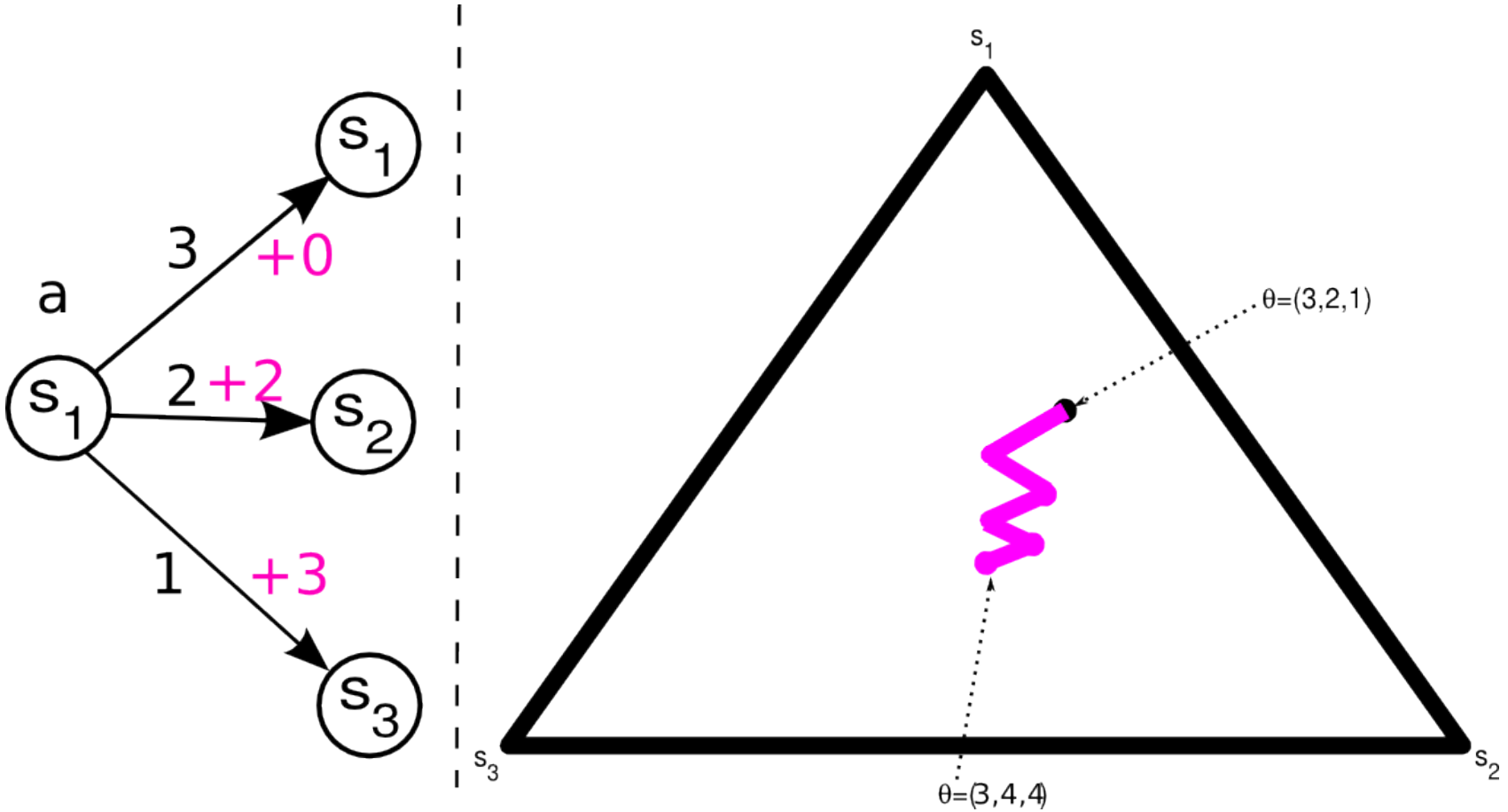




Bayes Boost $\sim$ Adding Artificial Trajectories

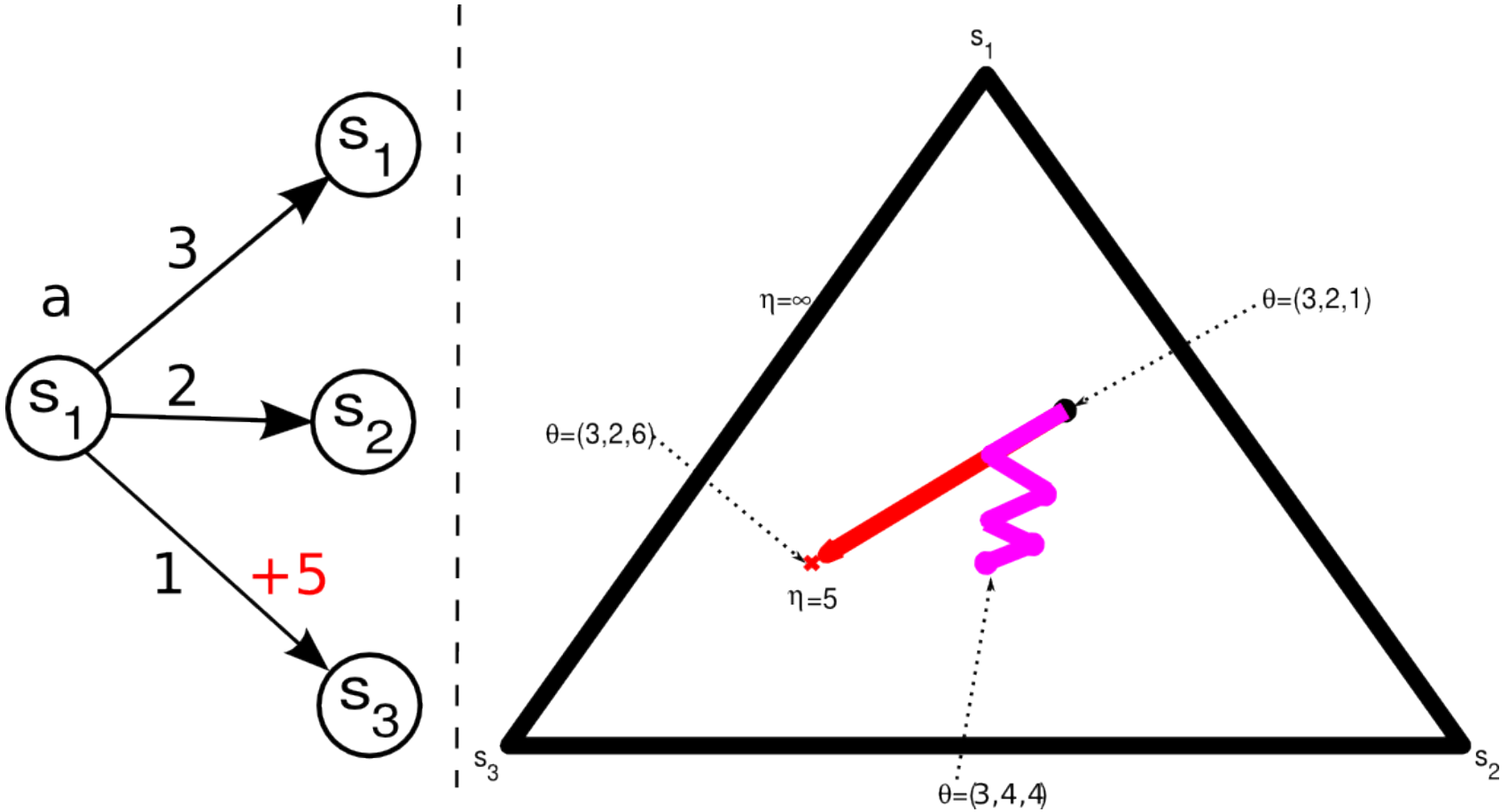


Bayes Boost $\sim$ Adding Artificial Trajectories



Bayes update of a trajectory $\mathbf{b}^h = Bayes(\mathbf{b}, s, a, h)$

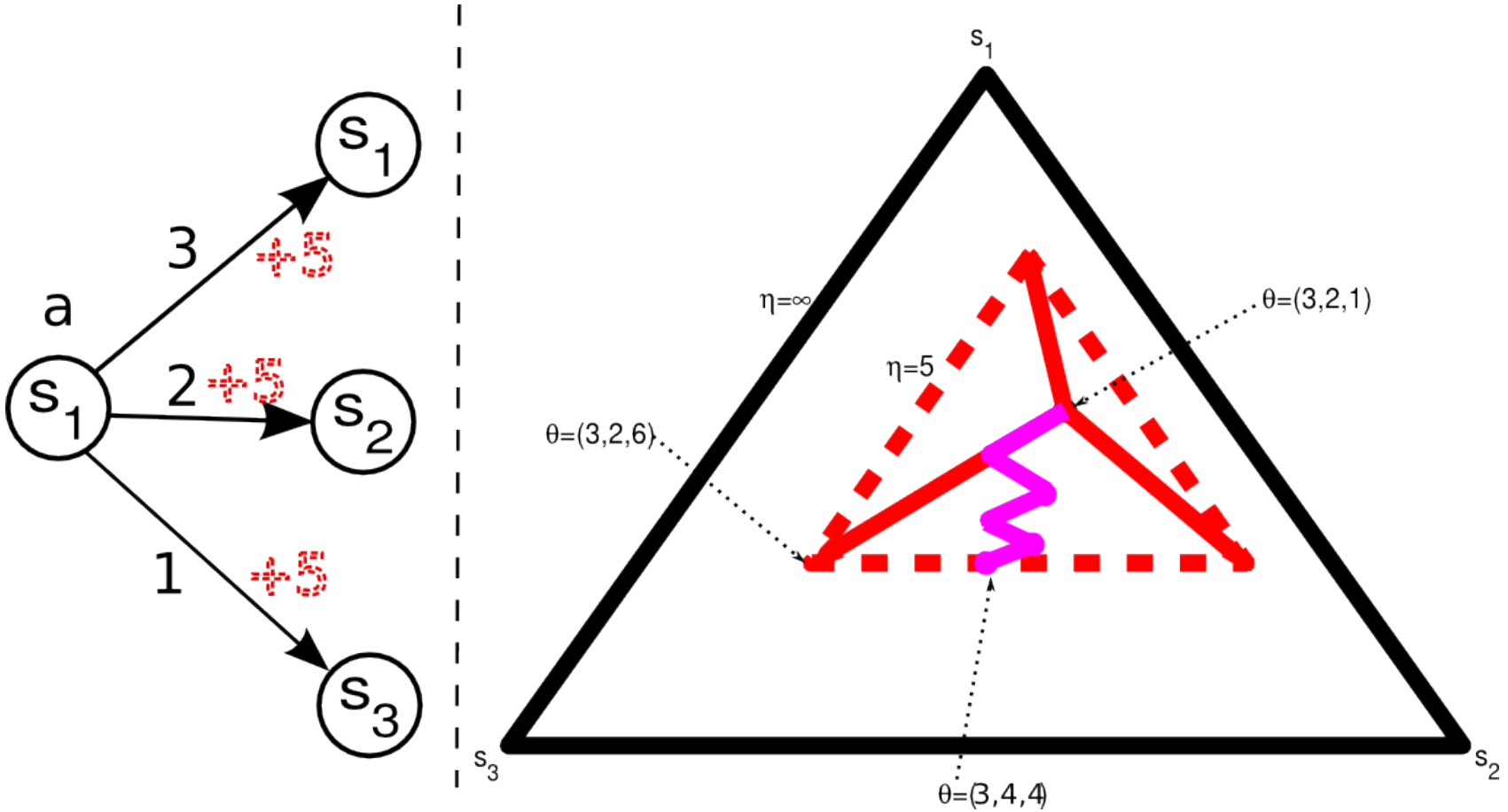
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η-Bayes boost $\mathbf{b}^{\sigma,\eta} = Bayes(\mathbf{b}, s, a, \sigma^\eta)$

Bayes Boost $\sim$ Adding Artificial Trajectories



Bayes update of a trajectory $\mathbf{b}^h = Bayes(\mathbf{b}, s, a, h)$

η-Bayes boost $\mathbf{b}^{\sigma,\eta} = Bayes(\mathbf{b}, s, a, \sigma^\eta)$

How to choose the σ direction? (for all state-action pairs)

BOLT $\sim$ Solving an Augmented-Action MDP

Augmented-Action MDP

Put σ in action space: $\alpha = \langle a, \sigma \rangle$

1	Update the belief	$\mathbf{b}' = Bayes(\mathbf{b}, s, a, s')$
2	Construct an η-Boosted transition function	$\hat{T}_\eta(s, \alpha, s')$
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Is a myopic but **optimistic** approximation if $\eta = H$

Lemma (Optimism)

Let $\hat{V}_H(s_t, \mathbf{b}_t)$ be BOLT's value iteration with an horizon of H and $\eta = H$ and let $\mathbb{V}_H(s_t, \mathbf{b}_t)$ be the optimal Bayesian value function. Then,

$$\hat{V}_H(s_t, \mathbf{b}_t) \geq \mathbb{V}_H(s_t, \mathbf{b}_t).$$

BOLT is PAC-BAMDP

Theorem (Approximately Correct with High Probability)

- *If* $\mathcal{A}_t$ *is the policy followed by BOLT at time* t,
- *and the optimism parameter is set to* $\eta = H$,
- *then, with probability at least* $1 - \delta$,
- *BOLT is* ϵ*-close to the optimal Bayesian policy*

$$\mathbb{V}^{\mathcal{A}_t}(s_t, \mathbf{b}_t) \geq \mathbb{V}^{*}(s_t, \mathbf{b}_t) - \epsilon$$

- *for all but* $\tilde{O}\left(\frac{|\mathcal{S}||\mathcal{A}|H^2}{\epsilon^2(1-\gamma)^2}\right)$ *time steps.*

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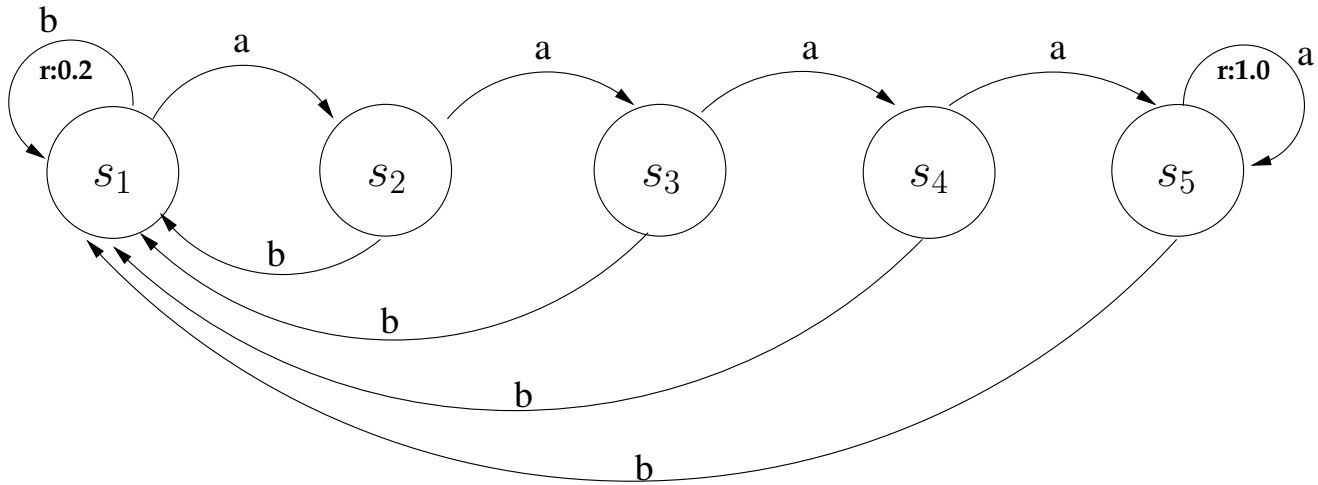
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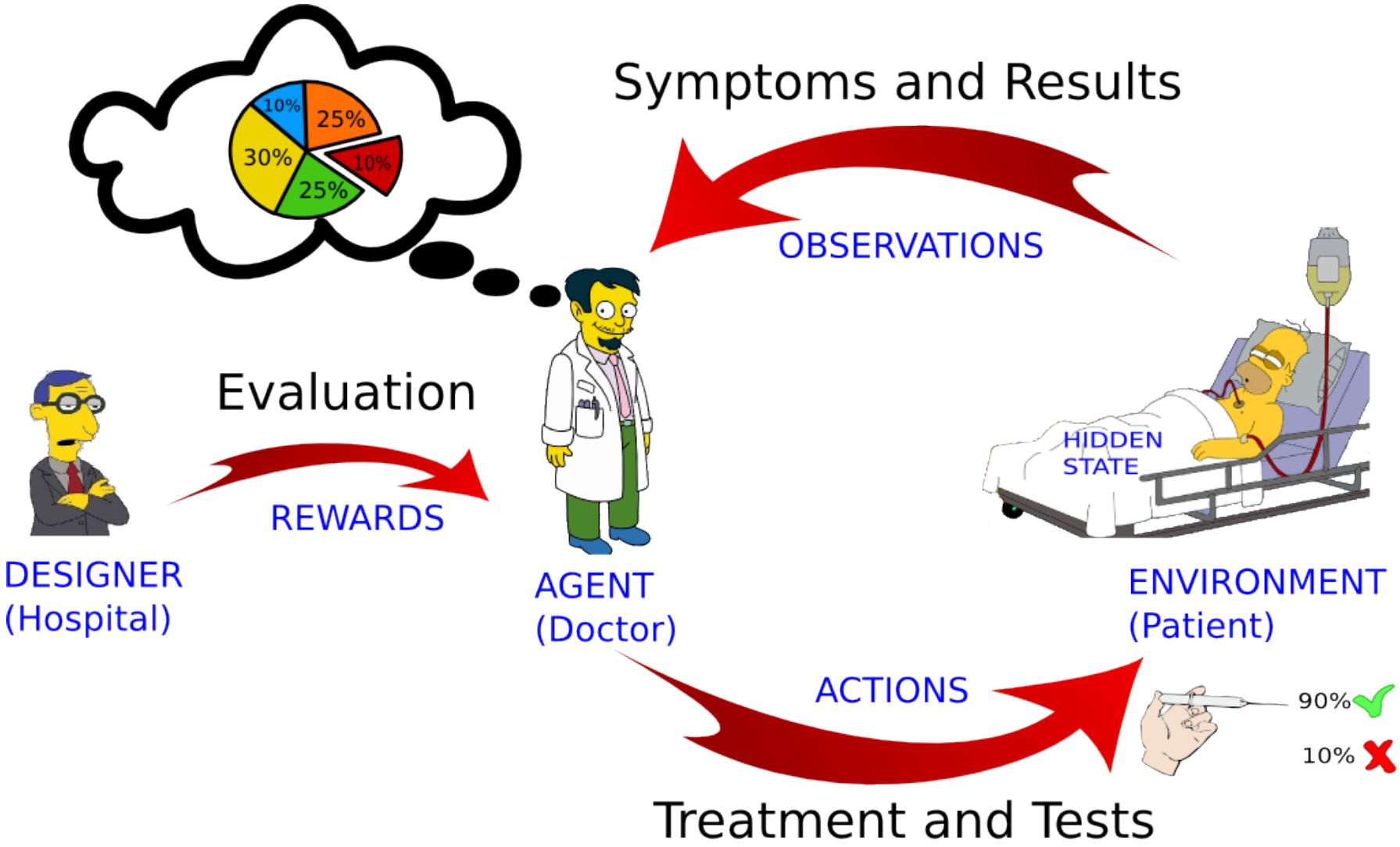
BEB bound $= \tilde{O}\left(\frac{|\mathcal{S}||\mathcal{A}|H^4}{\epsilon^2(1-\gamma)^2}\right)$, with $\beta = 2H^2$

Chain Problem (?)

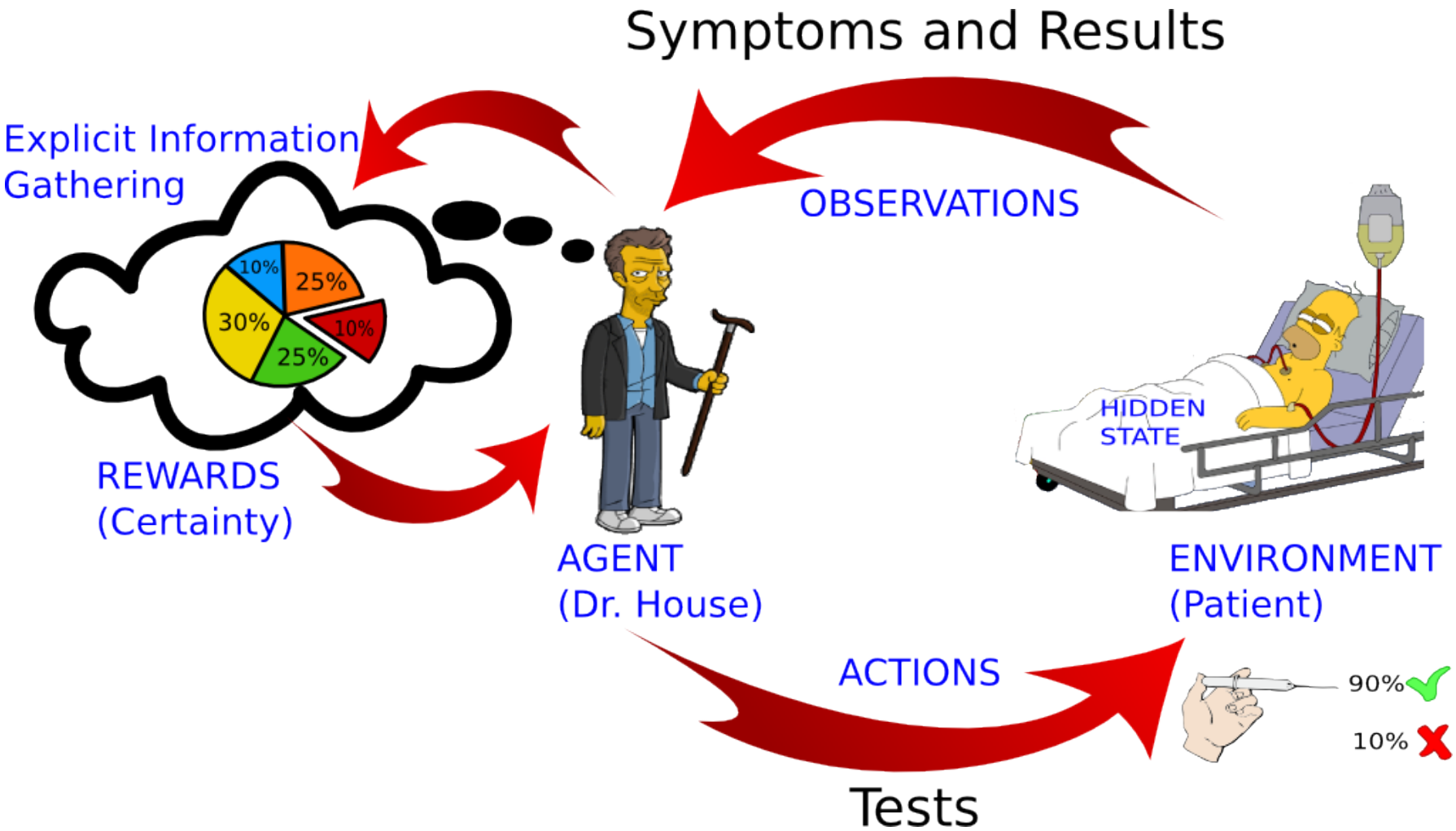


Algorithm	Mean Total Return
EXPLOIT ($\varepsilon = 0$)	230.2
BEB ($\beta = 1$)	**343.0**
BEB ($\beta = H$)	165.2
BOLT ($\eta = 7$)	289.6
BOLT ($\eta = H$)	278.7
BEETLE *	175.4
BOSS *	300.3

Explicit Information-Gathering in POMDPs

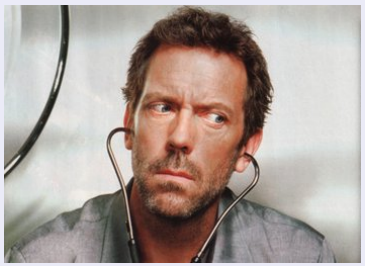


Explicit Information-Gathering in POMDPs



Explicit Information-Gathering Problems

Medical Diagnosis

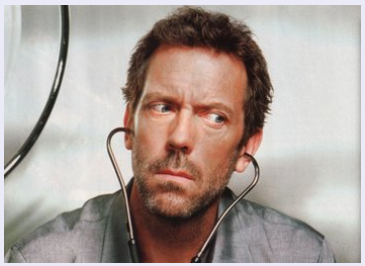


Doctor House

- Acquire information to disambiguate a disease
- Rewarded by its own knowledge

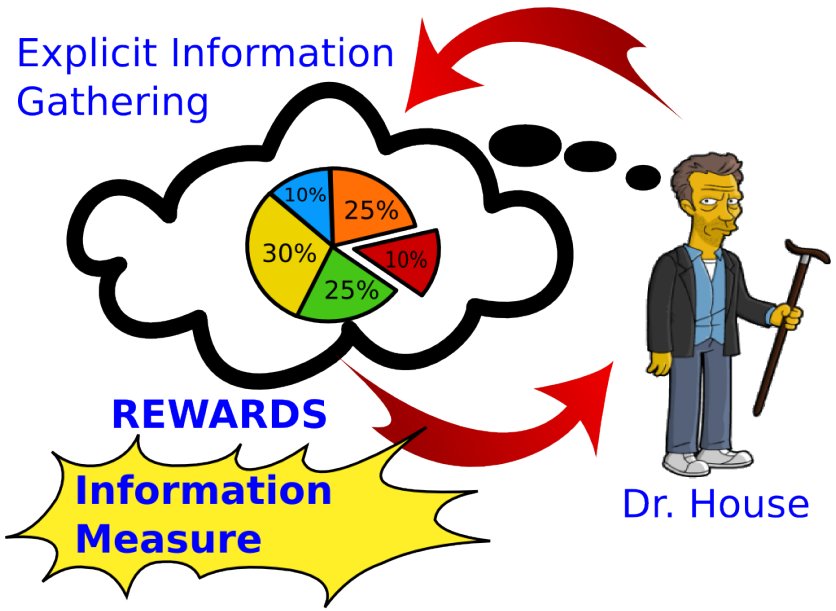
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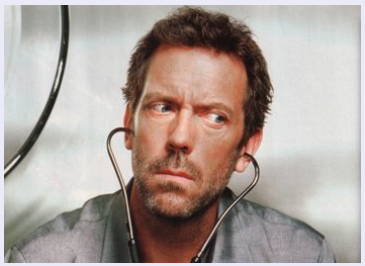
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- Needs an information measure (information theory)



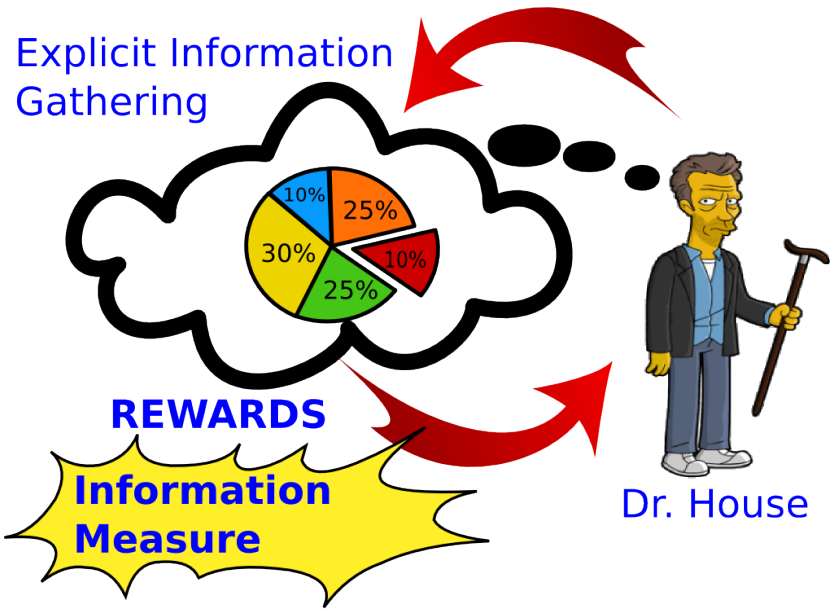
Explicit Information-Gathering Problems

Medical Diagnosis



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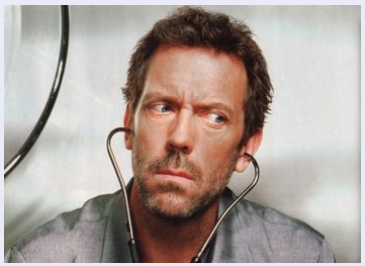
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- Military Surveillance
- Mapping and Active SLAM
- Resource monitoring
- Mobile Robot Localization
- etc...

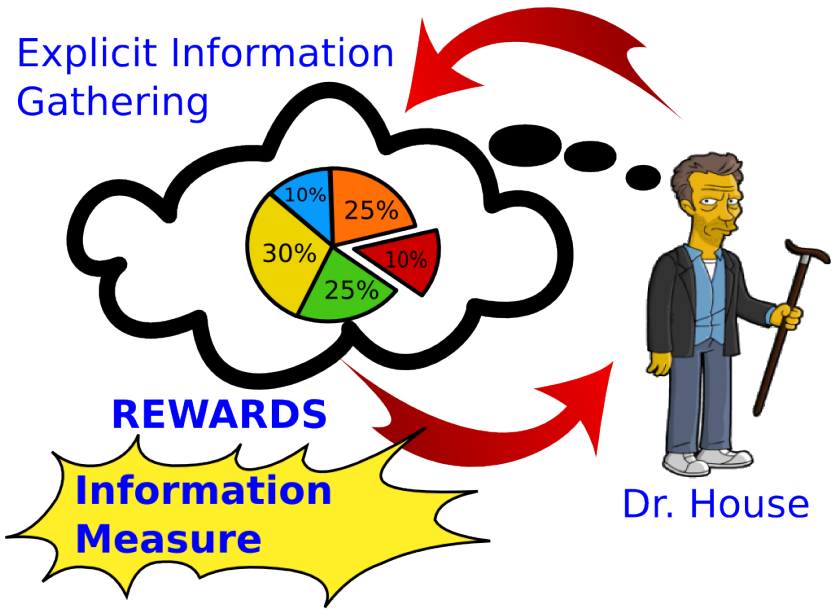
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POMDPs are not suitable for modelling Explicit IGPs

POMDP Extension for Explicit IGPs (ρPOMDP)

- $\rho(b, a) = \sum_{s \in \mathcal{S}} b(s) R(s, a)$ is not an information measure

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- ρPOMDP is a relaxation of POMDP

POMDP	**ρPOMDP**
$\langle \mathcal{S}, \mathcal{A}, \mathcal{Z}, T, O, R, b_0 \rangle$	$\langle \mathcal{S}, \mathcal{A}, \mathcal{Z}, T, O, f(b), b_0 \rangle$
$\downarrow$	$\downarrow$
belief MDP	**belief MDP**
$\langle \Delta, \mathcal{A}, \tau, \rho = \sum_s b(s) R(s,a), b_0 \rangle$	$\langle \Delta, \mathcal{A}, \tau, \rho = f(b), b_0 \rangle$

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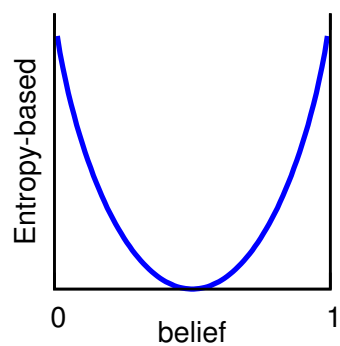
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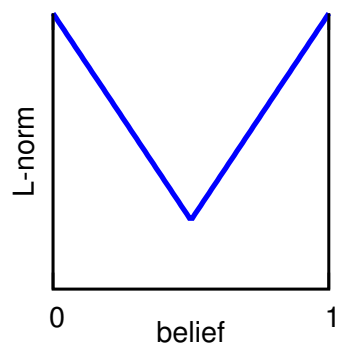
- Examples of ρ functions for IGPs



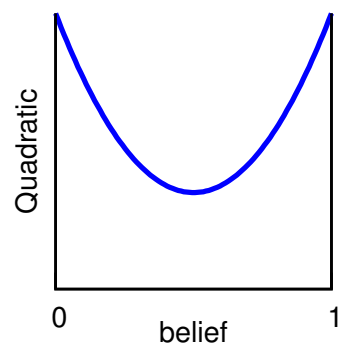
$\sum_s \mathbf{b}(s) R(s,a)$



$D(\mathbf{b} \| \mathbb{U})$



$\|\mathbf{b}\|_\infty$



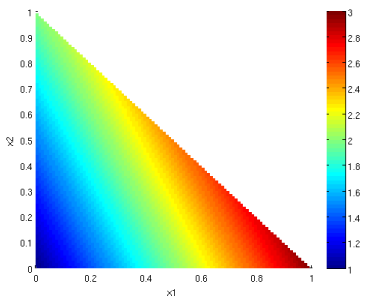
$\|\mathbf{b}\|_2^2$

POMDP Extension for Explicit IGPs (ρPOMDP)

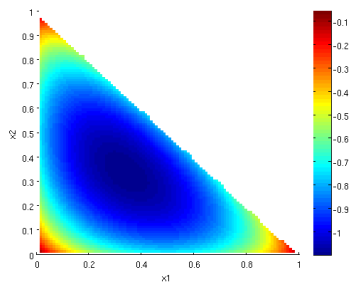
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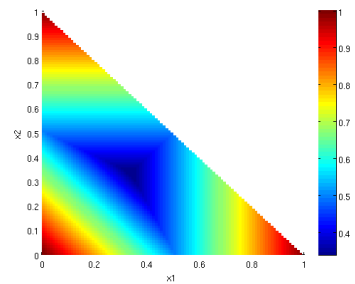
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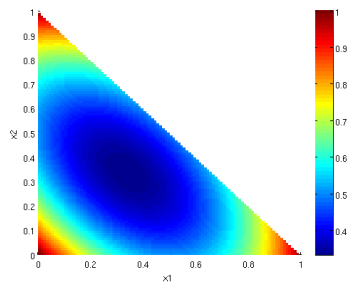
$\sum_s \mathbf{b}(s) R(s,a)$



$D(\mathbf{b} \| \mathbb{U})$

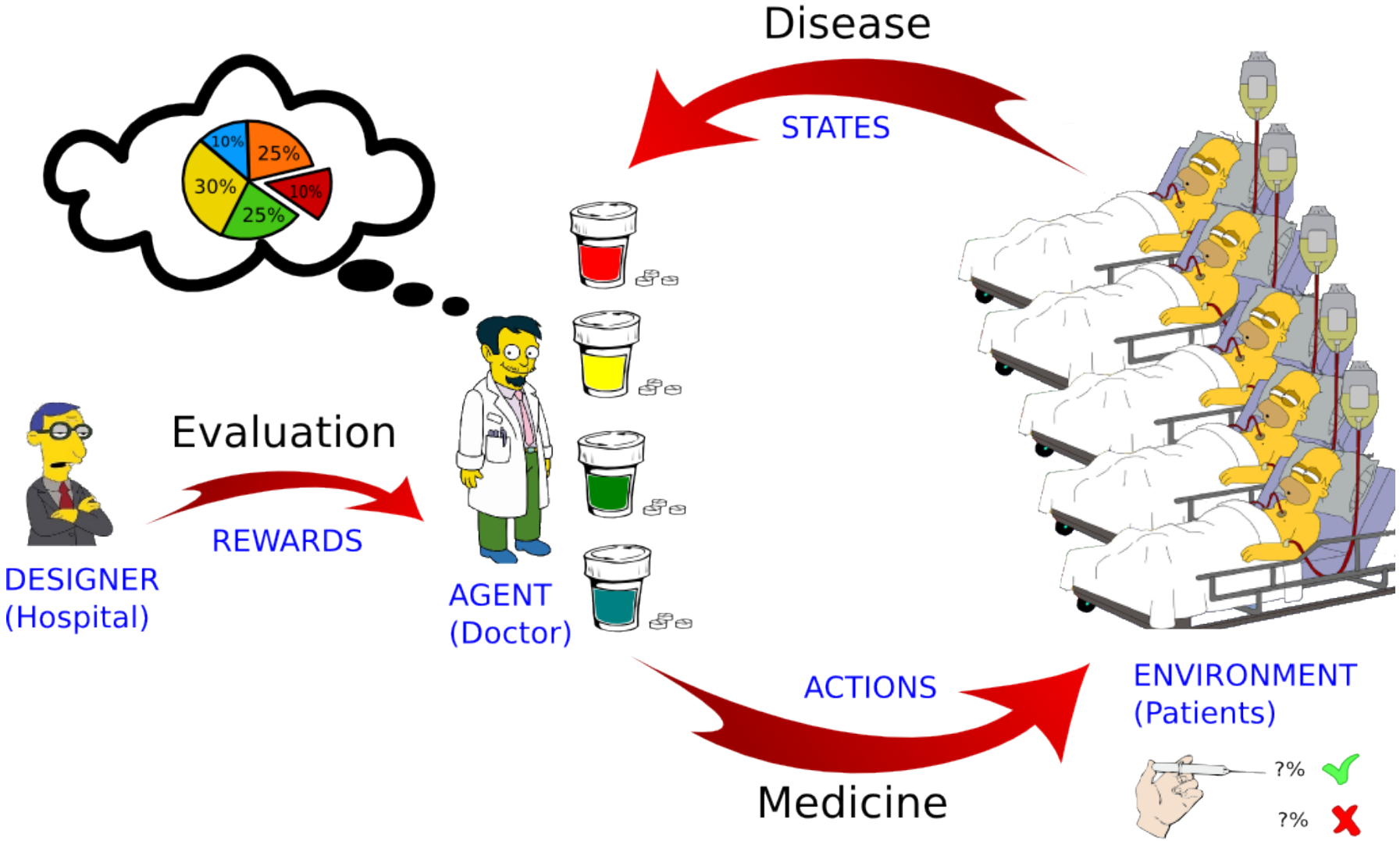


$\|\mathbf{b}\|_\infty$

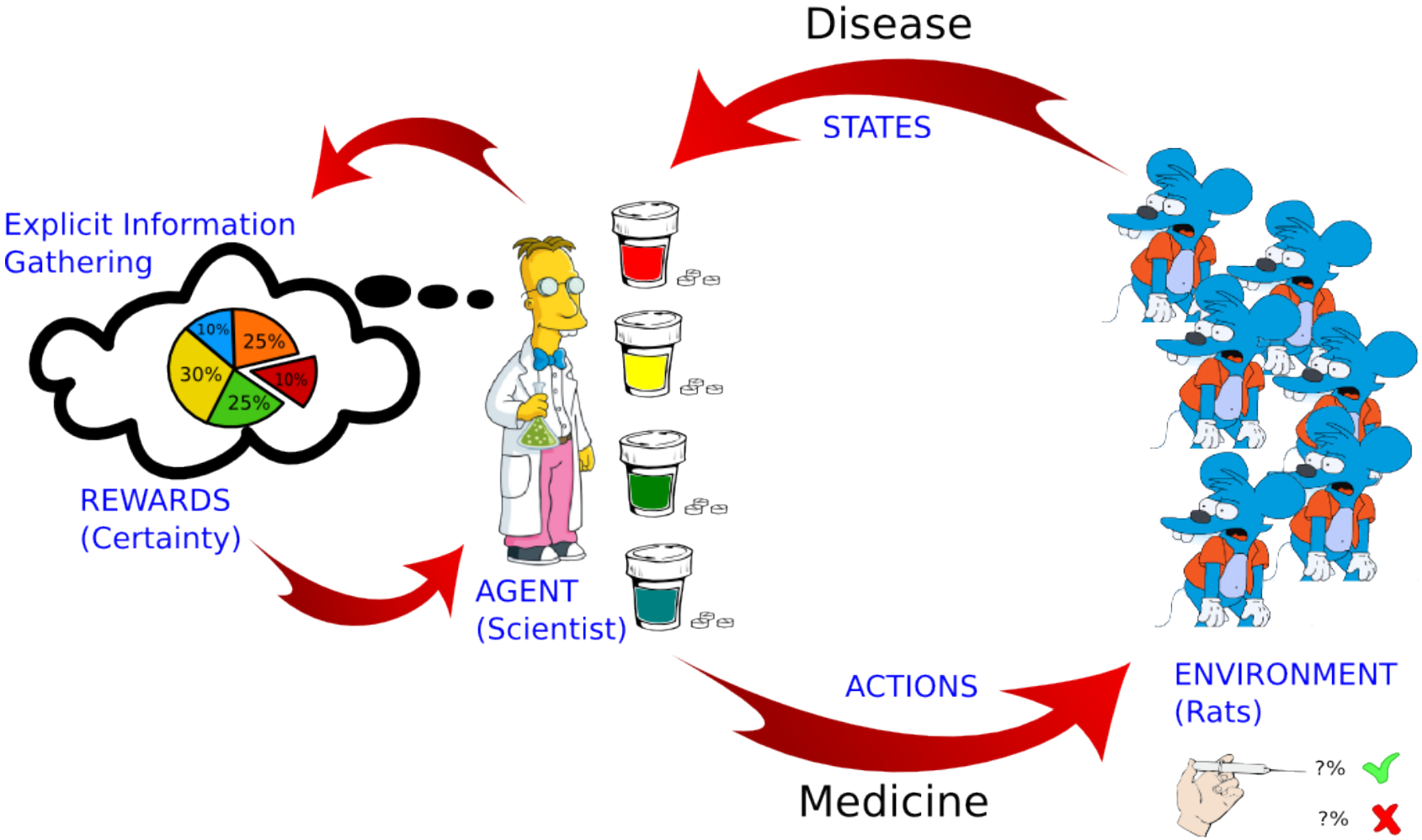


$\|\mathbf{b}\|_2^2$

Bayesian RL for Optimal Exploration



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Found a way to **approximately** solve explicit IGPs